

Linear-Time, Constant-Depth Blind Polynomial Commitments from Generalized RAA Codes, with an End-to-End Blind SNARK Implementation

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Abstract. We construct the first *blind polynomial commitment scheme* (PCS) whose prover performs strictly linear $O(N)$ work at constant FHE multiplicative depth, and we compose it into an end-to-end *blind SNARK* for arbitrary R1CS. We give what is, to our knowledge, the first *end-to-end measured* implementation of such a blind SNARK at scale, which we provide as supplementary material.

The construction rests on the *Generalized RAA code* over a fixed non-binary field $\mathbb{F}_q$, which encodes in linear time at constant homomorphic depth. We prove linear minimum distance in the fixed- q regime required by FHE plaintext moduli. The key estimate is a pairwise-product upper bound for the accumulator’s normalized input–output weight enumerator (IOWE), which drives the failure probability to $o(1/N)$ for every fixed prime power $q \geq 3$ (Lemma 2). Plugging the code into a Liger-style proximity test and composing it with a depth-controlled Spartan sum-check PIOP yields the blind SNARK, with $O(N)$ ciphertext–ciphertext multiplications and $O(N \log N)$ total FHE operations at constant depth. In our performance evaluation, we instantiate the scheme with $(q, N, k, \delta) = (2^{16} + 1, 2^{22}, 4, 0.30)$. Conditional on a successful code setup, the IOPP reaches 128-bit per-proof soundness with query count $\ell^* = 843$, and the end-to-end prover produces a proof in ~ 25 s. At $N = 2^{20}$ our single-core prover is over $400\times$ faster than PHALANX [30] and over $650\times$ faster than LAMINATE’s [23] fastest single-instance variant (Section 7). Neither prior scheme has released source code, whereas our implementation enables reproducible single-thread measurement on commodity hardware.

Keywords: Blind PCS · Blind SNARK · Verifiable FHE · Polynomial Commitment Schemes · Linear-Time Encoding · RAA Codes

1 Introduction

Verifiable Computation over Encrypted Data (VCoED) [14,9] composes fully homomorphic encryption (FHE) with succinct non-interactive arguments

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(SNARKs): a server homomorphically evaluates a circuit on a client’s encrypted input, and additionally produces a proof that the evaluation was performed correctly. When the proof itself is generated entirely over FHE ciphertexts, so that the client only ever sees encrypted intermediate state, the resulting object is a *blind* SNARK [23].

Two efficiency axes are fundamental for any practical blind SNARK [23]:

- **Prover complexity:** the number of FHE operations the prover performs as a function of the circuit size n .
- **Multiplicative depth:** the depth of homomorphic multiplications in the prover’s circuit, which directly controls FHE parameter blowup (and hence ciphertext size, evaluation noise budget, and bootstrapping frequency).

In leveled BGV/BFV [10,11], a depth- d prover requires ciphertext modulus $\log Q = \Theta(d \cdot \log q_0)$, where q_0 is the per-level prime. Each homomorphic operation on a ring- $\mathbb{F}_q[X]/(X^M+1)$ ciphertext then costs $\tilde{O}(M \cdot \log Q) = \tilde{O}(M \cdot d)$ via NTT-based polynomial arithmetic. Hence T depth- d ciphertext operations incur total work $T \cdot \tilde{O}(M \cdot d)$, giving a $\text{poly}(d)$ multiplicative blowup per ciphertext operation over the plaintext operation count.¹

The PCS-layer trilemma. A SNARK follows the standard modular blueprint of a polynomial IOP (PIOP) composed with a polynomial commitment scheme (PCS). Recent blind and FHE-friendly constructions, including Laminate [23], Phalanx [30], and the blind-SNARK line of [15], have concentrated most of their effort on the PIOP layer. The PCS layer presents a separate obstruction. Codes with usable distance guarantees have historically required either super-linear encoding or $\Omega(\log n)$ multiplicative depth, making PCS encoding the dominant homomorphic cost in NTT-based blind SNARKs.

We ask whether a blind PCS can simultaneously provide $O(n)$ prover work, $O(1)$ FHE multiplicative depth, and support for the non-binary plaintext modulus used by BGV/BFV. No prior construction attains all three. Once this trilemma is resolved, the bottleneck shifts: in our implementation, the Spartan PIOP sumcheck rather than the PCS dominates the measured prover time (Section 7). Existing code families miss at least one requirement:

- **Reed–Solomon codes** (used by Phalanx [30] and the RS-based variants of FRI [7], BaseFold [29], DeepFold [18], WHIR [5]). A 1D-NTT runs in $O(n \log n)$ time at $O(\log n)$ FHE depth. HELIOPOLIS [4] introduced the low-depth 2D-NTT evaluation of Reed–Solomon encoding; Phalanx [30] uses this decomposition to reduce depth to $O(1)$ at the cost of $O(n^{1+1/k})$ work.
- **Expander-based codes** (Brakedown [16], Orion [27]). Linear-time encoding, but each expander cascade contributes one FHE multiplicative level, so the FHE depth is $\Omega(\log n)$. The concurrent QA construction [22] uses an $O(\log n)$ -stage Walsh–Hadamard transform with the same depth cost.

¹ This BGV/BFV scaling is the standard accounting (see e.g. [10, §3]). Modulus-switching keeps $\log Q$ at $\Theta(d \log q_0)$ rather than letting noise grow exponentially.

Table 1. Codes for blind PCS. “Provable distance regime” is the parameter range in which a published proof gives a non-vacuous failure bound. Here n is the block length, and q, p are base-field sizes. The Generalized RAA code is shared with Akhiani–Zhang [2]. Their bound contains an $O(1/(q-1))$ term, whereas ours does not.

Code Family	Encoding	Mult. Depth	Field	Provable Distance Regime
Reed–Solomon (1D-NTT) [7]	$O(n \log n)$	$O(\log n)$	Non-binary	Any q
Reed–Solomon (2D-NTT) [4,30]	$O(n^{1+1/k})$	$O(1)$	Non-binary	Any q
Brakedown [16]	$O(n)$	$O(\log n)$	Non-binary	Any q
EA Code [8]	$O(n \log n)$	$O(1)$	Non-binary	Any q
QA Code [22]	$O(n \log n)$	$O(\log n)$	Non-binary	$p \gg n$, vacuous at fixed p
Binary RAA [12]	$O(n)$	$O(1)$	Binary	$q = 2$ only
Generalized RAA (this work)	$O(n)$	$O(1)$	Non-binary	Any fixed $q \geq 3$

- **Binary RAA [12].** Linear-time encoding at depth 1 with a tight distance proof, but restricted to $\mathbb{F}_2$, incompatible with the non-binary plaintext modulus used by standard FHE parameter sets (e.g. the Fermat prime $q = 2^{16} + 1$).

Where each candidate code fails. Table 1 compares candidate codes along the three axes that matter for blind PCS deployment: linear-time encoding, $O(1)$ multiplicative depth, and a provable distance bound at the FHE plaintext modulus. Reed–Solomon, Brakedown, and EA codes encounter the encoding-time versus depth tradeoff. QA codes have no asymptotic distance guarantee at fixed p , while binary RAA is restricted to $\mathbb{F}_2$.

The Generalized RAA code was also studied by Akhiani–Zhang [2]. Their failure-probability bound retains an $O(1/(q-1))$ term, which is too large at the FHE plaintext modulus (approximately $10^{-4.8}$ for $q = 2^{16} + 1$). Our fixed- q analysis removes this term and proves the required distance property over non-binary $\mathbb{F}_q$. We also give a setup-time parameter-selection procedure for fixed tuples (Section 5) and a full end-to-end implementation (Section 7).

1.1 Our Contributions

We give three contributions. First, we close the fixed- q distance analysis for Generalized RAA codes. Second, we construct the first blind PCS with $O(N)$ prover work at $O(1)$ FHE depth. Third, we give the first end-to-end *measured* blind SNARK implementation at $N = 2^{22}$ by composing this PCS with a depth-controlled Spartan PIOP.

Contribution 1: Distance analysis of Generalized RAA over fixed non-binary fields (Section 4.3). The Generalized RAA code already has the desired encoding complexity (linear encoding at depth 1 over a non-binary field), but a fixed- q distance theorem with explicit failure decay was missing. We supply it:

Theorem 1 (Linear Minimum Distance, informal). *For any fixed prime power $q \geq 3$ there exist a repetition threshold $K \in \mathbb{N}^+$ and a relative-distance*

constant $\delta > 0$ such that for $k > K$,

$$\Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} (\mathbb{F}_q \setminus \{0\})^N}} [\exists x \neq 0 : \text{wt}(\text{RAA}(x)) \leq \delta N] = o\left(\frac{1}{N}\right).$$

The threshold K may depend on q , while δ may be chosen universally. The asymptotic decay rate $o(1/N)$ holds for every fixed prime power $q \geq 3$.

For an accumulator, let $\tilde{A}(w, h)$ denote the probability that a uniformly randomized weight- w input produces an output of weight h . The proof rests on the following pairwise-product upper bound (Lemma 2):

$$\tilde{A}(w, h) \leq \frac{w^2}{N} \cdot \left(\frac{h}{N-w}\right)^{\lfloor (w-1)/2 \rfloor} \cdot 6^w.$$

The inequality follows from direct factorial bounds on the IOWE summands. Unlike prior analyses, it does not partition those summands according to their nonzero-symbol factor. It therefore contributes no additive $O(1/(q-1))$ term to the final failure bound. In contrast, the bound of [2] contains such a term, which does not vanish as $N \rightarrow \infty$. Our bound tends to zero for every fixed $q \geq 3$.

Contribution 2: First blind PCS with $O(N)$ work at $O(1)$ FHE depth (Section 6). Plugging the Generalized RAA code into a Liger-style proximity-test framework [3] yields a blind PCS with $O(N)$ FHE operations at constant depth. The protocol uses two public linear-combination queries, denoted `LinComb`, in which the prover combines committed rows using verifier-supplied coefficients. SIMD ciphertext repacking consolidates each response into a single packed ciphertext without increasing depth (Section 6.4). Brakedown-FHE has linear work but $\Omega(\log N)$ depth, while RS-based PCSs require either super-linear work or $O(\log n)$ depth.

Contribution 3: End-to-end blind SNARK implementation (Sections 6.2, 7). Composing the PCS with a Spartan-style sumcheck PIOP for R1CS yields a complete blind SNARK. To preserve $O(1)$ FHE depth, each round recomputes its partial fold from the original ciphertexts rather than applying `mult_plain` to the result of the previous round. This adds a $\log N$ factor in plaintext-ciphertext multiplication work. The resulting prover performs $O(N)$ ciphertext-ciphertext multiplications and $O(N \log N)$ total FHE operations at constant depth. Our Rust implementation uses Microsoft SEAL on a single Apple M4 Pro core, with BFV plaintext modulus $q = 2^{16} + 1$ and ring degree 2^{14} . It produces a 128-bit-soundness proof in approximately 25 seconds at $N = 2^{22}$. Section 7 gives the full breakdown.

Underlying tool: efficient concrete parameter selection (Section 5). For fixed inputs (N, q, k, δ) , RAA-CERTIFY efficiently computes a rigorous upper bound on the distance-failure probability. An explicit bound covers every possible output

weight of the second accumulation, and a two-state moment-generating matrix bounds the output-weight distribution of the first. The algorithm partitions the complete ranges $1 \leq h \leq N$ and $1 \leq w \leq N/k$ into intervals; it does not assume that either $D(h)$ or $T(w)$ is monotone. RAA-PARAMSELECT then searches candidate (k, δ) grids and returns the lowest-cost tuple satisfying a deployment-specified setup-failure budget. At $(N, q, k, \delta) = (2^{20}, 65537, 4, 0.30)$, the current prototype completes the ordinary floating-point pass in 24.70 seconds and gives the candidate unfiltered value $\log_{10} \epsilon_{\text{cert}} \approx -11.0304$. Our core concrete observation is that the weight-one contribution dominates this value. A deployment may either choose a conservative tuple whose complete unfiltered bound meets its budget, or perform an exact setup-time scan of every weight-one message and use a more aggressive tuple. At the displayed tuple, this optional scan leaves the analysis of all weights $w \geq 2$ and gives the filtered candidate $\log_{10} \epsilon_{\text{setup}} \approx -21.0817$. Final numerical verification requires directed-rounding interval arithmetic.

Two distinct budgets govern deployment (Section 5.6). *Soundness* against an adaptive prover is set by the IOPP query count ℓ and extension-field degree e , independently of the code-distance analysis. *Setup-time risk* is the one-time probability that the sampled code misses relative distance δ . A parameter tuple is used only when Algorithm 1 returns a machine-verified upper bound below the deployer’s chosen setup budget. Brakedown [16, §1.2, §7] likewise treats code generation as a public setup event. Our selector leaves the precise setup threshold to the deployment rather than identifying it with per-proof soundness.

1.2 Technical Overview

Blind PCS and SNARK construction. Over leveled BGV/BFV with plaintext field $\mathbb{F}_q$ and SIMD batch size B , an encrypted vector $\llbracket f \rrbracket \in \mathbb{F}_q^N$ is arranged as an $r \times m$ matrix. The rows are encoded and the resulting matrix is Merkle-committed column-wise, with B rows packed in each ciphertext. Evaluation and proximity testing use the LinComb operation defined above. SIMD repacking compresses each response (Section 6.4). The PCS performs $O(N)$ FHE operations at depth 1. Composing it with Spartan’s two-sumcheck PIOP gives an end-to-end SNARK with $O(N)$ ciphertext–ciphertext multiplications and $O(N \log N)$ total FHE operations at constant depth.

Distance analysis: stratified union bound. After repetition, a weight- w message enters the first randomized accumulation with weight kw . Let $\tilde{A}(kw, s)$ be the probability that this step produces intermediate weight s , and let $D(s)$ be the probability that the next randomized accumulation produces fewer than δN nonzero symbols. A union bound gives

$$\Pr[\text{dist}(C) < \delta N] \leq \sum_{w=1}^{N/k} \binom{N/k}{w} (q-1)^w \sum_{s=1}^N \tilde{A}(kw, s) D(s).$$

For $w \geq 0.01N/k$, entropy estimates obtained from Stirling’s formula give an exponentially small contribution (Theorem 3). For smaller w , the direct pairwise-product bound of Lemma 2 gives the required inverse-polynomial estimate.

Organization. After preliminaries (§2–3), §4 defines the code and states the distance theorem. Section 4.3 proves it, and §5 gives the failure-probability certifier and concrete parameter-search procedure. Section 6 constructs the blind PCS and SNARK. Section 7 evaluates performance.

2 Related Work

Our work intersects three threads: distance analysis of accumulator-based codes, blind polynomial commitments and verifiable FHE, and coding-based polynomial commitments more broadly. We briefly position the paper in each thread here. Appendix G gives the detailed technical comparisons.

Distance analysis of accumulator-based codes. The binary case ($q = 2$) has been studied for over two decades and is settled by [13, 21, 12]. Non-binary analyses in the $n \gg q$ regime [28, 20, 25, 17] introduce the code family and give approximate or numerically verified spectral analyses. They do not give a closed-form failure probability with explicit constants for a target (N, q, k, δ) . Akhiani–Zhang [2] prove $\tilde{O}(N^{-\eta}) + O(1/(q-1))$, which is tight when q grows super-polynomially but retains a non-vanishing term for fixed q . The concurrent Quasi-Abelian codes of [22] attain near-GV concrete distance for $p \gg N$. As their authors note, they have no asymptotic distance guarantee for fixed p , and their bound becomes vacuous at an FHE plaintext modulus. Lemma 2 addresses the fixed non-binary- q regime and yields failure probability $o(1/N)$ without a q -dependent residual term.

Blind PCS and verifiable FHE. Verifiable computation over encrypted data was framed by [14, 9] and instantiated with proximity-test IOPs by HELIOPOLIS [4]. HELIOPOLIS introduced the low-depth 2D-NTT evaluation of Reed–Solomon encoding, which Phalanx [30] uses to obtain $O(1)$ FHE depth with an $O(N^{1+1/k})$ prover. Laminate [23] obtains $O(N \log N)$ prover work at $O(1)$ depth through blind GKR and transcript packing. Lasagne [31] follows the same GKR and sum-check route, introducing “multiplication-layered” circuits to reflect FHE depth. Both GKR-based schemes are restricted to layered payload circuits. Phalanx, Laminate, and Lasagne report projected prover times obtained from operation counts and microbenchmarks rather than end-to-end measurements.

A parallel line of work [1] applies non-binary RAA codes in the plaintext domain for outsourced SNARKs, but does not address the constant-depth requirement created by encryption. Our PCS is the first to combine $O(N)$ prover work with $O(1)$ FHE depth. Composing it with a depth-controlled Spartan PIOP gives an end-to-end blind SNARK with $O(N \log N)$ total work at $O(1)$ depth, while supporting arbitrary R1CS rather than layered circuits.

Coding-based PCS more broadly. Reed–Solomon based schemes (FRI [7], BaseFold [29], Liger [3], DeepFold [18], WHIR [5]) pay $O(n \log n)$ encoding and $O(\log n)$ FHE multiplicative depth via NTT. Linear-time expander-based codes

(Brakedown [16], Orion [27]) cut encoding to $O(n)$ but reintroduce $O(\log n)$ depth from the expander cascade. QA codes [22] use $O(\log n)$ -stage Walsh–Hadamard transforms, also $O(\log n)$ depth. Binary RAA [12] is depth-1 and linear-time but field-restricted to $\mathbb{F}_2$. Table 1 (Introduction) summarises the encoding-time, multiplicative-depth, and provable-distance-regime axes.

Concurrent work further improves the plaintext coding-based PCS frontier. Lightning [24] composes a base code with a new linear code to trade relative distance for faster commitment. Bolt [19] develops sketched codes and code switching for fast multilinear commitments. ERA codes [6] combine encoding, repetition, and accumulation with an IOPP for field-agnostic hash-based SNARKs. These results are complementary to our setting: they optimize plaintext hash-based commitments, whereas we analyze fixed-modulus distance and homomorphic encoding depth for a blind PCS over FHE ciphertexts.

The detailed survey in Appendix G expands each thread. In particular, it compares our direct combinatorial upper bound with the partitioned IOWE analysis of [2], and it gives a more detailed comparison with [1].

3 Preliminaries

For $x \in \mathbb{F}_q^k$, let $\text{wt}(x)$ denote its Hamming weight. We use ρ and δ for relative Hamming weights, C for a linear code or its generator matrix, and $\mathcal{C}$ for a code ensemble. We write encoding as the row-vector product $c = xC$.

3.1 Coding Theory

Definition 1 (Input-Output Weight Enumerator). For an $[n, k]$ code C , its input-output weight enumerator (IOWE) is

$$A_{w,h}(C) = |\{x \in \mathbb{F}_q^k : \text{wt}(x) = w \text{ and } \text{wt}(xC) = h\}|.$$

Thus $A_{w,h}(C)$ counts weight- w messages whose encodings have weight h .

Definition 2 (Code Ensemble and Average IOWE). A code ensemble is a collection $\mathcal{C} = \{C_r\}$ indexed by a random seed r . Its average IOWE is

$$\bar{A}_{w,h}(\mathcal{C}) = \frac{1}{|\mathcal{C}|} \sum_{C \in \mathcal{C}} A_{w,h}(C)$$

Definition 3 (Accumulate Code). The accumulate code A is the memory-1, rate-1 convolutional code with generator matrix

$$A = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

Thus $(x_1, \dots, x_n)A = (x_1, x_1 + x_2, \dots, x_1 + \dots + x_n)$.

Definition 4 (Repeat Code). The rate- $1/k$ repeat code R has generator matrix

$$R = \begin{pmatrix} R' & & \\ & R' & \\ & & \ddots \\ & & & R' \end{pmatrix} \in \mathbb{F}_q^{n \times kn}$$

where $R' = (1 \ 1 \ \dots \ 1) \in \mathbb{F}_q^k$. It repeats each symbol of x exactly k times.

3.2 Polynomial Commitment Schemes and IOPPs

Definition 5 (Polynomial Commitment Scheme). A polynomial commitment scheme (PCS) for μ -variate multilinear polynomials over $\mathbb{F}_q$ is a tuple of algorithms (Setup, Commit, Eval):

- **Setup**($1^\lambda, \mu$) $\rightarrow$ **pp**: given security parameter λ and number of variables μ , outputs public parameters.
- **Commit**(**pp**, $\tilde{f}$) $\rightarrow$ ($C, \mathcal{D}$): commits to a multilinear polynomial $\tilde{f} : \mathbb{F}_q^\mu \rightarrow \mathbb{F}_q$ and returns a commitment C together with prover auxiliary data $\mathcal{D}$.
- **Eval**(**pp**, $C, \mathbf{z}, y; \tilde{f}, \mathcal{D}$): an interactive protocol between prover $\mathcal{P}$ and verifier $\mathcal{V}$ in which $\mathcal{P}$ convinces $\mathcal{V}$ that the committed polynomial satisfies $\tilde{f}(\mathbf{z}) = y$ on a verifier-chosen evaluation point $\mathbf{z} \in \mathbb{F}_q^\mu$.

The scheme must satisfy the following three security properties.

1. **Completeness.** For every μ -variate multilinear $\tilde{f}$, every $\mathbf{z} \in \mathbb{F}_q^\mu$, and $y = \tilde{f}(\mathbf{z})$, an honest execution of **Eval**(**pp**, $C, \mathbf{z}, y; \tilde{f}, \mathcal{D}$) on $(C, \mathcal{D}) \leftarrow \text{Commit}(\text{pp}, \tilde{f})$ has the verifier accept with probability 1 over the protocol's internal randomness.
2. **Binding.** For every PPT adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} \tilde{f}_0 \neq \tilde{f}_1 \wedge \\ \text{Eval}(\text{pp}, C, \mathbf{z}_0, y_0; \tilde{f}_0, \mathcal{D}_0) = \text{accept} \wedge \\ \text{Eval}(\text{pp}, C, \mathbf{z}_1, y_1; \tilde{f}_1, \mathcal{D}_1) = \text{accept} \end{array} \middle| \begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda, \mu); \\ (C, \{(\tilde{f}_b, \mathcal{D}_b, \mathbf{z}_b, y_b)\}_{b \in \{0,1\}}) \leftarrow \mathcal{A}(\text{pp}) \end{array} \right] \leq \text{negl}(\lambda).$$

No PPT adversary, given **pp**, can produce a single commitment C together with two convincing openings for distinct polynomials, except with negligible probability.

3. **Knowledge Soundness.** There exists a PPT extractor $\mathcal{E}$ such that, for every PPT prover $\mathcal{P}^*$ that causes **Eval** to accept on commitment C with non-negligible probability ϵ for some $(\mathbf{z}, y)$, $\mathcal{E}^{\mathcal{P}^*}(\text{pp}, C)$ outputs $\tilde{f}$ satisfying $\tilde{f}(\mathbf{z}) = y$ with probability at least $\epsilon - \text{negl}(\lambda)$.

Definition 6 (Interactive Oracle Proof of Proximity (IOPP)). Let $C \subseteq \mathbb{F}_q^n$ be a linear code with relative minimum distance δ . An IOPP for C is an interactive protocol $\langle \mathcal{P}, \mathcal{V} \rangle$ in which $\mathcal{V}$ has oracle access to a purported codeword $\mathbf{y} \in \mathbb{F}_q^n$ and to the messages $\mathcal{P}$ writes onto a public oracle:

1. Completeness. If $\mathbf{y} \in C$, the verifier interacting with the honest $\mathcal{P}$ accepts with probability 1.
2. Soundness (proximity gap). If $\mathbf{y}$ is more than $\delta/3$ -far from C (in relative Hamming distance), then for any (possibly unbounded) malicious prover $\mathcal{P}^*$, the verifier accepts with probability at most $\beta(\delta, \ell)$, where ℓ is the verifier's query complexity and β is the protocol's soundness error.

For the Ligerio proximity-test framework [3] we use, the soundness error takes the form $\beta(\delta, \ell) = (1 - \delta/3)^\ell + \text{negl}(\lambda)$ where the negligible term comes from a Schwartz–Zippel check on a verifier-chosen random linear combination.

3.3 (Leveled) Homomorphic Encryption

We use a leveled homomorphic encryption (LHE) scheme [11,10] $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ over plaintext space $\mathcal{P} = \mathbb{F}_p^D$, with standard syntax and a maximum multiplicative depth MAXDEPTH beyond which decryption is no longer guaranteed correct. BGV/BFV ciphertexts are SIMD-packed across D slots [10]: an encryption of $\mathbf{m} \in \mathbb{F}_p^D$, denoted $[\![\mathbf{m}]\!]$, supports component-wise homomorphic addition $[\![\mathbf{u}]\!] + [\![\mathbf{v}]\!]$, plaintext–ciphertext multiplication $\mathbf{w} \cdot [\![\mathbf{u}]\!]$, and ciphertext–ciphertext multiplication $[\![\mathbf{u}]\!] \cdot [\![\mathbf{v}]\!]$, all acting on all D slots in parallel. Throughout this paper, $D \in \{2^{13}, 2^{14}\}$ (the standard BFV/BGV SIMD batch sizes).

4 Generalized RAA Codes and Distance Analysis

We introduce the code family, state the main distance theorem, and give the proof.

Notation (Sections 4–5). Throughout the distance analysis, N denotes the *code-word block length* of the (per-row) Generalized RAA code and N/k its message length. In the blind-PCS application (Sections 6 onward) the witness is arranged in a balanced Ligerio matrix and *each row* is encoded by this code, so a code-word length N supports a circuit of size $(N/k)^2$; the application sections write the circuit size itself as N and the per-row codeword length explicitly.

Definition 7 (Random Interleavers [21]). A uniformly sampled permutation $\pi \in S_n$ induces the rate-1 code P_π defined by $xP_\pi = (x_{\pi(1)}, \dots, x_{\pi(n)})$ for $x = (x_1, \dots, x_n) \in \mathbb{F}_q^n$.

Definition 8 (Random Weightors). For $\mathbf{r} = (r_1, \dots, r_n)$ sampled uniformly from $(\mathbb{F}_q \setminus \{0\})^n$, the random weightor $W_{\mathbf{r}}$ acts as $xW_{\mathbf{r}} = (r_1x_1, \dots, r_nx_n)$.

Definition 9 (Generalized RAA Code Ensemble). The Generalized RAA code encodes $x \in \mathbb{F}_q^{N/k}$ as

$$c = xR(P_{\pi_1}W_{\mathbf{r}_1}A)(P_{\pi_2}W_{\mathbf{r}_2}A),$$

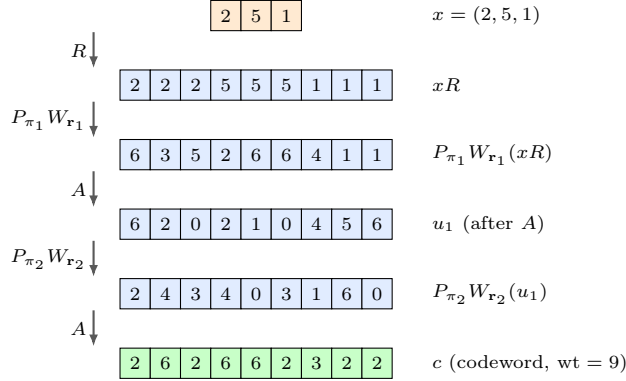


Fig. 1. The Generalized RAA encoding pipeline of Example 1 ($q = 7$, $k = 3$, $x = (2, 5, 1)$, all arithmetic mod 7): repeat (R), interleave-and-weight ($P_{\pi_1} W_{r_1}$), accumulate (A), and a second interleave-weight-accumulate round. Every stage is an element-wise multiplication or a prefix-sum scan (no field inversions or FFTs), so the pipeline is natively FHE-compatible.

where R is the rate-1/ k repeat code, P_{π_i} are random permutations, W_{r_i} are random diagonal weightors, and A is the accumulator (prefix-sum operator). The **ensemble** $\mathcal{C}_k$ is the collection over all (π_1, π_2, r_1, r_2) .

Example 1 (RAA Encoding with $k = 3$). For $q = 7$, $N/k = 3$, $k = 3$, $x = (2, 5, 1)$ and parameters $\pi_1 = (1, 4, 7, 2, 5, 8, 3, 6, 9)$, $r_1 = (3, 2, 5, 1, 4, 6, 2, 3, 1)$, $\pi_2 = (5, 9, 2, 7, 3, 8, 1, 4, 6)$, $r_2 = (2, 3, 5, 1, 4, 2, 6, 3, 1)$, the pipeline produces the codeword $c = (2, 6, 2, 6, 6, 2, 3, 2, 2)$ (all arithmetic mod 7); Figure 1 traces the intermediate vectors stage by stage. The codeword c has Hamming weight 9 (all nonzero); every step is element-wise multiplication or a prefix-sum scan, with no field inversions or FFTs, so the pipeline is natively FHE-compatible.

The key structural properties are immediate:

- **Linear-time encoding.** Each operation (repeat, permute, scale, accumulate) is $O(N)$.
- **Constant multiplicative depth.** Per FHE-operation accounting: the repeat step R is a plaintext rearrangement (depth 0); the permutation π_i is a public coordinate reshuffle implemented either via SIMD slot rotations (depth 0) or, when packed across B rows, by a permutation network of constant-depth rotate-and-mask blocks (which we account as depth 0.5 to flag the masking pt-ct multiplication); the weightor W_{r_i} is one round of plaintext-ciphertext multiplication by a public weight vector (depth 0.5, since pt-ct mults consume half the noise budget of a ct-ct mult); the accumulator A is the depth-0 prefix-sum scan implemented with ct-ct additions only. Summing along the pipeline $R \rightarrow (P_{\pi_1} W_{r_1}) \rightarrow A \rightarrow (P_{\pi_2} W_{r_2}) \rightarrow A$, the total multiplicative depth is $0.5 + 0.5 = 1$ level, one pt-ct multiplication's worth of noise growth from each weightor, accumulated additively.

- **Defined over any prime power.** The encoding pipeline is well-defined over any finite field $\mathbb{F}_q$ with $q > 2$; the distance analysis (Section 4.3) covers the fixed- q regime.

4.1 IOWE Machinery

The input-output weight enumerator (IOWE) of the non-binary accumulator is [28,25,17]:

$$A_{w,h} = \sum_{i=0}^{w-1} \binom{h-1}{\lceil \frac{w-i}{2} \rceil - 1} \binom{N-h}{\lfloor \frac{w-i}{2} \rfloor} \cdot \binom{h - \lceil \frac{w-i}{2} \rceil}{i} (q-1)^{\lceil \frac{w-i}{2} \rceil} (q-2)^i.$$

Lemma 1 (IOWE of composed encoders [21]). *For two codes C_1, C_2 over $\mathbb{F}_q$, where C_1 has block length n , the average IOWE of the ensemble $\mathcal{C} = \{C_1 P_\pi W_{\mathbf{r}} C_2\}$ is $\bar{A}_{w,h}(\mathcal{C}) = \sum_{l=1}^n \frac{A_{w,l}(C_1) A_{l,h}(C_2)}{\binom{n}{l} (q-1)^l}$.*

The random interleaver makes the composition memoryless on the support distribution, while the random weightor eliminates the dependence on specific nonzero values. If we composed two codes directly without an interleaver, the weight distribution of the final codeword would depend on both the support pattern and the values of the nonzero elements in the intermediate codeword, yielding a correlated distribution. The random interleaver breaks this correlation by uniformly randomizing the support, while the weightor uniformly randomizes the values. As a result, the IOWE of the composed code factors cleanly into a product of the individual IOWEs, the key structural property that enables our analysis.

4.2 Main Theorem

Theorem 2 (Linear Minimum Distance, Asymptotic). *For any fixed $q > 2$, there exist $K \in \mathbb{N}^+$ and $\delta > 0$ such that for repetition factor $k > K$:*

$$\Pr_{\substack{\pi_1, \pi_2 \xleftarrow{\$} S_N \\ \mathbf{r}_1, \mathbf{r}_2 \xleftarrow{\$} (\mathbb{F}_q \setminus \{0\})^N}} [\exists x \neq 0 : \text{wt}(\text{RAA}(x)) \leq \delta N] = o\left(\frac{1}{N}\right).$$

The threshold K may depend on q but not on N ; the distance constant δ may be chosen universally. The asymptotic conclusion $o(1/N)$ holds for any fixed prime power $q \geq 3$.

Remark 1 (Explicit dependence on q). The following gives one explicit sufficient choice. Put $\eta_0 = 0.01$ and choose

$$\delta_0 := \frac{0.99}{144e^2}.$$

Let $\psi_{\eta_0} := \Psi_{\eta_0}((1 - \eta_0)/199) < 0$ be the exponent from Appendix D.1, set $\Lambda_{\eta_0} := -\psi_{\eta_0}/2$, and put

$$\mu_0 := \eta_0 \Lambda_{\eta_0}, \quad a_0 := 6\sqrt{\frac{2\delta_0}{0.99}}, \quad b_0 := 6\sqrt{a_0}\sqrt{\frac{\eta_0}{2(1 - \eta_0)}}.$$

Here e is Euler's number, $a_0 < e^{-1}$, and $b_0 < 1$. One sufficient threshold is

$$K(q) := \left\lceil \max \left\{ 9, \frac{\log q}{\mu_0}, \frac{\log(e(q-1)/\eta_0)}{-\log b_0} \right\} \right\rceil. \quad (1)$$

Then the theorem holds with $\delta = \delta_0$ for every fixed $k > K(q)$ and all sufficiently large N . The term involving μ_0 closes the large-weight branch. Indeed, $\eta_0 > 2\delta_0$, so the accumulator's support condition makes the second-accumulator collapse probability exactly zero whenever its input weight exceeds $\eta_0 N$. The constants 9 and b_0 close the two endpoints of the remaining small-weight branch. Floors, ceilings, and polynomial prefactors affect only the threshold on N .

4.3 Distance Proof

We prove Theorem 2 by bounding the probability that any nonzero input x maps to a codeword of weight below δN .

Proof strategy: stratified union bound. A naïve union bound over all q^N possible nonzero inputs would be exponentially loose. Instead, we partition the input space by weight and bound each weight class separately. Write the normalized IOWE $\tilde{A}(w, h) = A_{w,h}/[\binom{N}{w}(q-1)^w]$ and, for a relative threshold η , define

$$D_\eta(s) := \sum_{h=1}^{\lfloor \eta N \rfloor} \tilde{A}(s, h), \quad D(s) := D_\delta(s).$$

Using the union bound and stratifying by input weight:

$$\begin{aligned} \Pr[\text{dist}(C) < \delta N] &= \Pr[\exists x \neq 0 : \text{wt}(\text{RAA}(x)) \leq \delta N] \\ &\leq \sum_{w=1}^{N/k} \Pr[\exists x : \text{wt}(\text{RAA}(x)) \leq \delta N \mid \text{wt}(x) = w]. \end{aligned} \quad (2)$$

Weight evolution through the pipeline. For a fixed input x of weight w , we trace the weight distribution through the encoding process $c = xR(P_{\pi_1}W_{\mathbf{r}_1}A)(P_{\pi_2}W_{\mathbf{r}_2}A)$:

1. *Repeater:* The output of xR has weight exactly kw , since each nonzero coordinate is replicated k times.
2. *Random interleaver + weightor:* Since P_{π_1} and $W_{\mathbf{r}_1}$ preserve the number of nonzero coordinates, the weight remains kw after these operations.

3. *First accumulator*: The weight distribution of $xRP_{\pi_1}W_{\mathbf{r}_1}A$ is governed by the (kw) -th row of the IOWE matrix of the accumulator, i.e., $\Pr[\text{wt}(xRP_{\pi_1}W_{\mathbf{r}_1}A) = s] = \tilde{A}(kw, s)$.
4. *Second round*: Similarly, for the second round, $\Pr[\text{wt}(xP_{\pi_2}W_{\mathbf{r}_2}A) = h \mid \text{wt}(x) = s] = \tilde{A}(s, h)$.

Composing via Lemma 1, the probability that a weight- w input produces a low-weight output is:

$$\Pr[\exists x : \text{wt}(\text{RAA}(x)) \leq \delta N \mid \text{wt}(x) = w] \leq \left(\frac{N}{k}\right)^w (q-1)^w \sum_{s=1}^N \tilde{A}(kw, s) \cdot D(s). \quad (3)$$

Our objective is to show that each weight- w term in (2) is $o(1/N^2)$, so that summing over all $w \in [N/k]$ yields $o(1/N)$. The analysis splits at $w = 0.01N/k$ into two regimes, each requiring distinct analytical techniques. The constant 0.01 is a convenient source of slack, not an empirically estimated phase transition. It makes the repeated weight in the large branch at least $0.01N$, while in the small branch it gives $N/(N-kw) \leq 100/99$. Any sufficiently small fixed constant could replace 0.01, with corresponding changes to the proof constants.

Large Weight Regime ($w \geq 0.01N/k$).

Intuition. In this regime the input weight to the first accumulator is $kw \geq 0.01N$, but its output weight remains random. We split according to whether that output falls below $0.01N$. Such a collapse is exponentially unlikely by applying the following theorem with threshold $\eta_0 = 0.01$; above the split, the second accumulator's low-output probability is exponentially small by applying it with threshold δ .

Theorem 3 (Exponential decay of $D(w)$). *Fix any η in the regime where the function Ψ_η below is monotone decreasing on its relevant range. There exist constants $\Lambda_\eta > 0$ and N_η such that, for every $N \geq N_\eta$ and every $r \geq 0.01N$,*

$$D_\eta(r) \leq e^{-\Lambda_\eta r}.$$

Proof sketch. Bound $\tilde{A}(r, h)$ for $h \leq \eta N$ by replacing the nonzero-symbol factor $(q-1)^a(q-2)^i/(q-1)^r$ by 1. Entropy estimates obtained from Stirling's formula then bound the remaining trinomial sum. Writing $a = \alpha N$, $h = \beta N$, and $i = \theta N$ gives $\tilde{A}(r, h) \leq rN^{O(1)} \max_i e^{\Phi(\alpha, \beta, \theta)N}$, where the polynomial factor also absorbs the one-unit parity gap between $\lceil (r-i)/2 \rceil$ and $\lfloor (r-i)/2 \rfloor$. Maximizing first over β and then over θ reduces the exponent to $\Psi_\eta(\alpha)$. In the stated regime, $\psi_\eta := \Psi_\eta((1-\eta)/199) < 0$. Thus $D_\eta(r) \leq rN^{O(1)} e^{\psi_\eta N}$. Since $N \geq r$ and $\psi_\eta < 0$, we have $\psi_\eta N \leq \psi_\eta r$; the polynomial prefactor is at most $e^{\varepsilon r}$ for all sufficiently large N and any fixed $\varepsilon > 0$. Taking $0 < \varepsilon < -\psi_\eta$ and $\Lambda_\eta = -\psi_\eta - \varepsilon$ proves the theorem. The full derivation, including the entropy estimate, the $\partial\Phi/\partial\beta$ and $\partial\Phi/\partial\theta$ calculations, the monotonicity check for Ψ , and the resulting δ range, is given in Appendix D.1.

Corollary 1. Fix δ in the regime of Theorem 3. There exist $K \in \mathbb{N}^+$ and $C > 0$ such that, for $k > K$, N sufficiently large, and $w \geq 0.01N/k$,

$$\binom{N/k}{w} (q-1)^w \sum_{s=1}^N \tilde{A}(kw, s) D_\delta(s) \leq e^{-CN}.$$

Corollary 2. Under the same assumptions, there exist $K \in \mathbb{N}^+$ and $C > 0$ such that, for $k > K$ and every $w \leq 0.01N/k$,

$$\binom{N/k}{w} (q-1)^w \sum_{s=\lceil 0.01N \rceil}^N \tilde{A}(kw, s) D_\delta(s) \leq e^{-CN}.$$

The first corollary applies Theorem 3 twice: $D_{0.01}(kw)$ controls collapse in the first accumulator, while $D_\delta(s)$ controls the second accumulator above the split. The second corollary uses only the latter bound. Full derivations appear in Appendix D.2.

Small Weight Regime ($w < 0.01N/k$).

Intuition: bounding all symbol assignments uniformly. When the input weight w is small, the Stirling-based bounds used in the large-weight regime become too loose: they estimate $o(1)$ but cannot achieve the $o(1/N^2)$ precision needed to survive the union bound over $O(N)$ weight classes. Prior analyses of accumulator-based codes [8, 2] partition the IOWE sum by collision count i . On one part they retain the nonzero-symbol factor $(q-1)^a (q-2)^i / (q-1)^w$, which contributes $O(1/(q-1))$ to the final bound. On the other part they replace this factor by the trivial upper bound 1. This decomposition is what produces the $O(1/(q-1))$ floor that prevents prior bounds from going below $\sim 10^{-5}$ when $q = 2^{16} + 1$.

We instead apply $\frac{(q-1)^a (q-2)^i}{(q-1)^w} \leq 1$ uniformly for every i . This removes the q -dependent term. We then bound the complete remaining combinatorial sum directly, using a pairwise-product inequality. For small input weight w and output weight $h \leq \delta N$, the resulting upper bound is sufficient to avoid the $O(1/(q-1))$ term in the prior analysis.

Lemma 2 (Pairwise-Product IOWE Bound). For $N \geq 100$, $w \leq 0.01N$, and $1 \leq h \leq N - w$:

$$\tilde{A}(w, h) \leq \frac{w^2}{N} \cdot \left(\frac{h}{N-w} \right)^{\lfloor (w-1)/2 \rfloor} \cdot 6^w$$

Consequently, for every $\delta \leq 0.30$:

$$D(w) \leq \frac{w^2}{\lfloor (w-1)/2 \rfloor + 1} \cdot \frac{\delta N + 1}{N} \cdot \left(\frac{\delta N + 1}{N-w} \right)^{\lfloor (w-1)/2 \rfloor} \cdot 6^w.$$

Proof sketch. The nonzero-symbol factor $(q-1)^a(q-2)^i/(q-1)^w$ is at most 1 because $a+i \leq w$. Write $a = \lceil (w-i)/2 \rceil$ and $b = \lfloor (w-i)/2 \rfloor$. Direct factorial estimates, together with $\binom{N}{w} = (N/w)\binom{N-1}{w-1}$ and $N/(N-w) \leq 100/99 < 2$, give a per-summand upper bound

$$\frac{w}{N} 6^w \left(\frac{h}{N-w} \right)^{\lfloor (w-1)/2 \rfloor}.$$

Summing first over i and then over $h \leq \delta N$ proves the two displayed bounds. Appendix D.3 gives the parity cases $a = b$ and $a = b + 1$.

Remark 2 (Regime split vs. [2]). The analysis of [2] partitions the IOWE sum at $i = w_2 - 2w_0 - 1$. It retains the nonzero-symbol factor on one part and uses the trivial bound on the other. This gives the asymptotically tight rate $\tilde{O}(N^{-(k-2)})$ when $q \gg n$, but it also introduces an additive $O(1/(q-1))$ term when q is fixed. Lemma 2 applies the trivial bound uniformly and controls the combinatorial part directly. The prior decomposition is tighter when $q = \omega(\text{poly}(n))$, because its nonzero-symbol factor decays as $1/(q-1)^{(w_1-i)/2}$. Ours is tighter for fixed q , where it has no q -dependent residual term. The FHE-plaintext regime ($q = 2^{16} + 1$, $q \ll N$) is in our favourable half of the trade.

Lemma 3. *There exists K such that for $k > K$ and $w \leq 0.01N/k$:*

$$\binom{N/k}{w} (q-1)^w \sum_{s=1}^{\lceil 0.01N \rceil - 1} \tilde{A}(kw, s) \cdot D(s) = o(1/N^2).$$

Proof sketch. Substituting the bounds from Lemma 2 for both $\tilde{A}(kw, s)$ and $D(s)$ gives a product whose logarithmic derivative in s is negative for sufficiently small δ . The largest summand is therefore attained at the left endpoint, up to constant slack. After applying $\binom{N/k}{w} \leq (Ne/(kw))^w$ and accounting for at most $0.01N$ summands, the per-weight bound has the form

$$p(w) = C_1 \cdot C_2^w \cdot w^{O(1)} \cdot (w/N)^{(k/2-1)w+O(1)}$$

for constants C_1, C_2 independent of (N, w) . This is enough to conclude $p(w) = o(1/N^2)$ uniformly over $w \leq 0.01N/k$ once k is sufficiently large and δ is sufficiently small. Full derivation, including the explicit thresholds on k and δ produced by the analysis, is in Appendix D.4.

Theorem 4 (Small-weight bound). *There exists K such that for $k > K$ and $w \leq 0.01N/k$:*

$$\binom{N/k}{w} (q-1)^w \sum_{s=1}^N \tilde{A}(kw, s) \cdot D(s) = o(1/N^2).$$

The theorem follows by splitting the inner sum at $s = 0.01N$ and bounding each half: the lower half by Lemma 3 ($o(1/N^2)$), the upper half by Corollary 2 ($e^{-\Omega(N)}$). See Appendix D.5.

Completing the Proof. The proof of Theorem 2 follows by summing (3) over w , then splitting at $w = 0.01N/k$: the large-weight portion is $e^{-\Omega(N)}$ by Corollary 1, the small-weight portion is $o(1/N^2)$ per term by Theorem 4, and summing over $w \in [N/k]$ gives $o(1/N)$. The full proof is in Appendix D.5.

5 Concrete Distance Analysis and Parameter Selection

Theorem 2 establishes asymptotic linear distance for fixed q and sufficiently large k and N . A PCS deployment instead fixes a finite block length and typically uses a small repetition factor, so we analyze each concrete tuple (N, q, k, δ) directly. We express its distance-failure probability as a union bound over all message weights and intermediate weights, majorize the second-accumulator failure probability by a geometrically contracting function, and bound the first-accumulator lower tail with a two-state generating matrix. Interval partitions then combine these bounds over the complete weight ranges. This gives the efficient RAA-CERTIFY procedure for evaluating a fixed tuple and a feasibility oracle for searching cost-minimizing parameters under a deployment-specified setup-failure budget.

Proof outline. A weight- w message has weight kw after repetition. To simplify the notation in the concrete analysis and certification procedure, let H_s denote the output weight of the first randomized accumulator when its input has weight s . Thus its intermediate output weight is H_{kw} . Writing $d = \lceil \delta N \rceil - 1$, recall the two central quantities

$$D(h) := \Pr[\text{wt}(yPWA) \leq d \mid \text{wt}(y) = h],$$

$$T(w) := \binom{N/k}{w} (q-1)^w \mathbb{E}[D(H_{kw})].$$

Thus $D(h)$ is the conditional failure probability of the second accumulator, and $T(w)$ is the union-bound contribution of all weight- w messages. To bound every $T(w)$, we first derive an explicit decreasing upper bound on $D(h)$. We then use a two-state generating matrix to bound the probability that H_{kw} lies in a prescribed weight interval. Summing over partitions of the complete h - and w -ranges gives the failure-probability upper bound returned by RAA-CERTIFY.

5.1 The Concrete Union Bound

Put

$$q_0 := q - 1, \quad n := N/k, \quad d := \lceil \delta N \rceil - 1.$$

Thus a codeword has relative weight below δ exactly when its weight is at most d . The case $d = 0$ has failure probability zero because every stage of the encoder is injective on nonzero inputs, so below we assume $d \geq 1$. For an accumulator whose randomized input has weight h , define

$$D(h) := \Pr[\text{wt}(yPWA) \leq d \mid \text{wt}(y) = h] = \sum_{s=0}^d \frac{A_{h,s}}{\binom{N}{h} q_0^h}. \quad (4)$$

Here PW is an independent random interleaver and weightor, and $A_{\cdot,\cdot}$ is the unnormalized accumulator IOWE from Section 3. Section 4.3 writes $\tilde{A}(s, h)$ for the corresponding normalized IOWE. For concise notation in the concrete analysis and certification procedure, we represent this distribution by the random output weight H_s . Hence

$$\Pr[H_s = h] = \tilde{A}(s, h) = \frac{A_{s,h}}{\binom{N}{s} q_0^s}. \quad (5)$$

The random interleaver chooses the support uniformly and the random weightor chooses every nonzero value independently and uniformly, so Equation (5) applies to the repeated word with $s = kw$.

For a message of weight w , let

$$T(w) := \binom{n}{w} q_0^w \mathbb{E}[D(H_{kw})]. \quad (6)$$

The standard union bound is then

$$\Pr[\text{dist}(C) < \delta N] \leq \sum_{w=1}^n T(w). \quad (7)$$

The following lemmas give an efficiently computable upper bound on the right-hand side.

5.2 An Explicit Upper Bound for the Second Accumulator

The first observation removes more than half of the possible intermediate-weight range.

Lemma 4. *For every $h > 2d$, $D(h) = 0$.*

For $0 \leq h \leq N$, define

$$U_{\star}(h) := \sum_{t=0}^{\min\{h,d\}} \frac{\binom{h}{t}^2 \binom{d}{t}}{\binom{N}{t}} q_0^{-(h-t)}. \quad (8)$$

Theorem 5 (Second-accumulator upper bound). *For every $0 \leq h \leq N$,*

$$D(h) \leq U_{\star}(h).$$

Moreover, for every $x \in (0, 1)$ and $0 \leq h < N$,

$$U_{\star}(h+1) \leq \left(\frac{1}{q_0(1-x)^2} + \frac{d/N}{x^2} \right) U_{\star}(h). \quad (9)$$

Consequently,

$$U_{\star}(h+1) \leq \rho_{\star} U_{\star}(h), \quad \rho_{\star} := \left(q_0^{-1/3} + (d/N)^{1/3} \right)^3. \quad (10)$$

In particular, $U_{\star}$ is globally decreasing when $\rho_{\star} < 1$.

Proof (Proof sketch). Condition on the h support positions of the accumulator input. Their gap vector is a uniformly random weak composition of $N - h$ into $h + 1$ bins. A union bound over the prefix indices at which the running sum returns to zero gives

$$D(h) \leq \frac{1}{\binom{N}{h}} \sum_{j=0}^h \binom{h}{j} \binom{N-h+j}{j} \binom{d}{h-j} q_0^{-j}.$$

Substituting $t = h - j$ yields Equation (8). For the contraction, write the t -th summand as $u_{h,t}$. The two exact parent ratios are

$$\frac{u_{h+1,t}}{u_{h,t}} = \frac{(h+1)^2}{q_0(h+1-t)^2}, \quad \frac{u_{h+1,t}}{u_{h,t-1}} = \frac{(h+1)^2(d-t+1)}{t^2(N-t+1)}.$$

Split the child terms at $t = \lfloor x(h+1) \rfloor$ and compare the two parts to the corresponding parents. This proves Equation (9). Minimizing its coefficient over x gives Equation (10). Appendix E.1 contains the full counting and endpoint details. $\square$

For the parameters $(q, N, \delta) = (65537, 2^{20}, 0.30)$, Equation (10) gives $\rho_\star < 0.3346$. This is a proved contraction factor for the explicit majorant $U_\star$.

5.3 A Two-State Moment-Generating Function for the First Accumulator

For $a \in (0, 1]$ and $z > 0$, define the positive matrix

$$M_a(z) := \begin{pmatrix} 1 & q_0 z a \\ z a(1 + (q-2)z) & \end{pmatrix}. \quad (11)$$

The states indicate whether the running accumulator value is zero or nonzero. The variable z marks a nonzero input coordinate, and a marks a nonzero output coordinate.

Lemma 5. *Let $e_0 = (1, 0)^T$ and $\mathbf{1} = (1, 1)^T$. For a polynomial $F(z)$, let $\text{coeff}_s(F)$ denote the coefficient of z^s . Then*

$$\text{coeff}_s(e_0^T M_a(z)^N \mathbf{1}) = \sum_{\substack{x \in \mathbb{F}_q^N \\ \text{wt}(x)=s}} a^{\text{wt}(Ax)}. \quad (12)$$

Consequently,

$$\mathbb{E}[a^{H_s}] = \frac{\text{coeff}_s(e_0^T M_a(z)^N \mathbf{1})}{\binom{N}{s} q_0^s}.$$

Proof. From state zero, a zero increment remains zero and each of the q_0 nonzero increments enters the nonzero state. From a nonzero state, a zero increment remains nonzero, exactly one nonzero increment returns to zero, and the other $q-2$ nonzero increments remain nonzero. These four possibilities are exactly the entries of (11). Multiplying over all coordinates and extracting the coefficient of z^s proves the claim. $\square$

Remark 3. The matrix is a transfer matrix for accumulator paths. Each multiplication advances one coordinate, while the powers of z and a record the input and output weights. The coefficient functional therefore isolates inputs of weight s exactly, and the positive-matrix comparison below converts this identity into an optimizable lower-tail bound.

For $r > 0$, put

$$\lambda(a, z, r) := \max \left\{ 1 + q_0 zar, a(1 + (q - 2)z) + \frac{z}{r} \right\}, \quad C(r) := \max\{1, 1/r\}. \quad (13)$$

Lemma 6. *For every $a \in (0, 1]$, $z, r > 0$, and integer R ,*

$$\Pr[H_s \leq R] \leq \frac{a^{-R} C(r) \lambda(a, z, r)^N}{z^s \binom{N}{s} q_0^s}. \quad (14)$$

Proof. For $v = (1, r)^T$, the definition of λ gives $M_a(z)v \leq \lambda(a, z, r)v$ component-wise, and $\mathbf{1} \leq C(r)v$. Therefore $e_0^T M_a(z)^N \mathbf{1} \leq C(r) \lambda(a, z, r)^N$. The nonnegative-coefficient inequality $\text{coeff}_s(F) \leq F(z)/z^s$ and $\mathbf{1}[H_s \leq R] \leq a^{H_s - R}$ complete the proof. $\square$

5.4 Complete Failure-Probability Certification

Theorem 6 (Complete failure-probability bound). *Assume $\rho_\star < 1$, and partition the complete active intermediate-weight range into consecutive intervals*

$$[1, H_\star] = \mathcal{B}_1 \sqcup \cdots \sqcup \mathcal{B}_J, \quad H_\star := \min\{N, 2d\}, \quad \mathcal{B}_j = [L_j, R_j].$$

For every $w \in [n]$, every $j \in [J]$, and arbitrary parameters $a_{w,j} \in (0, 1]$, $z_{w,j} > 0$, and $r_{w,j} > 0$,

$$T(w) \leq \binom{n}{w} q_0^w \sum_{j=1}^J U_\star(L_j) \frac{a_{w,j}^{-R_j} C(r_{w,j}) \lambda(a_{w,j}, z_{w,j}, r_{w,j})^N}{z_{w,j}^{kw} \binom{N}{kw} q_0^{kw}}. \quad (15)$$

Summing the displayed quantities over all $1 \leq w \leq n$ is therefore an upper bound on $\Pr[\text{dist}(C) < \delta N]$.

Proof. Lemma 4 removes $h > 2d$. On $\mathcal{B}_j$, Theorem 5 and $\rho_\star < 1$ give $D(h) \leq U_\star(h) \leq U_\star(L_j)$. Hence

$$\mathbb{E}[D(H_{kw})] \leq \sum_{j=1}^J U_\star(L_j) \Pr[H_{kw} \leq R_j].$$

Applying Lemma 6 to each term and multiplying by the outer factor in (6) proves (15). Finally, sum over the complete range $w \in [n]$ in (7). $\square$

We batch the outer sum over a consecutive message-weight interval $\mathcal{I} \subseteq [n]$. Fix one intermediate-weight interval $[L, R]$ and use a common parameter triple (a, z, r) for all $w \in \mathcal{I}$. The w -dependent factor in Equation (15) is

$$\Gamma_w(z) := \frac{\binom{n}{w} q_0^w}{z^{kw} \binom{N}{kw} q_0^{kw}} = \frac{\binom{n}{w}}{\binom{N}{kw}} (q_0^{1-k} z^{-k})^w. \quad (16)$$

Then the contribution of this pair of weight intervals is at most

$$U_\star(L) a^{-R} C(r) \lambda(a, z, r)^N \sum_{w \in \mathcal{I}} \Gamma_w(z). \quad (17)$$

The finite scalar sum in (17) is evaluated directly over every integer $w \in \mathcal{I}$. Sharing one parameter triple over $\mathcal{I}$ reduces the number of witness optimizations, while the complete message-weight and intermediate-weight partitions retain every term in (7).

Algorithm 1: RAA-CERTIFY: distance-failure probability certification.

Input: N, q, k, δ ; an optional exact set $\mathcal{W}_0$ and complete w - and h -interval partitions
Output: An upper bound ϵ_{cert} satisfying $\Pr[\text{dist}(C) < \delta N] \leq \epsilon_{\text{cert}}$, or FAIL

- 1 $q_0 \leftarrow q - 1$; $n \leftarrow N/k$; $d \leftarrow \lceil \delta N \rceil - 1$; $H_\star \leftarrow \min\{N, 2d\}$;
- 2 Compute an outward-rounded interval enclosure of $\rho_\star$;
- 3 **if** $\sup \rho_\star \geq 1$ **then**
- 4 **return** FAIL
- 5 Bound every $w \in \mathcal{W}_0$ by directed-rounded exact IOWE sums;
- 6 Verify that $\mathcal{W}_0$ and the w -intervals partition $[1, n]$;
- 7 Verify that the h -intervals partition $[1, H_\star]$;
- 8 **foreach** w -interval $\mathcal{I}$ and h -interval $[L, R] \subseteq [1, H_\star]$ **do**
- 9 Use floating point to choose a parameter triple (a, z, r) ;
- 10 Re-evaluate every factor in (17), including $U_\star(L)$, a^{-R} , $C(r)$, $\lambda(a, z, r)$, and $\sum_{w \in \mathcal{I}} \Gamma_w(z)$, with directed-rounding intervals;
- 11 **if** the parameter domain or interval evaluation cannot be verified **then**
- 12 | Refine the failing interval, or **return** FAIL;
- 13 Add the upper endpoint of (17) to ϵ_{cert} ;
- 14 **return** ϵ_{cert}

Numerical evaluation. Theorem 6 proves the bound returned by Algorithm 1, provided every arithmetic operation is enclosed by directed-rounding intervals. Ordinary floating point is used only to choose the interval partitions and the parameters (a, z, r) . The current prototype covers every $1 \leq w \leq N/k$ and completes the ordinary floating-point pass for $(N, q, k, \delta) = (2^{20}, 65537, 4, 0.30)$ in 24.70 seconds, giving the candidate value

$$\log_{10} \epsilon_{\text{cert}} \approx -11.0304,$$

with $w = 1$ dominating and the batched $w \geq 4$ contribution below $10^{-27.13}$. This value comes from the full-range calculation. It becomes a verified numerical bound once the same calculation is repeated with directed-rounding interval arithmetic.

Small-weight screening. Our core deployment observation is that $T(1)$ dominates the certified failure bound at the baseline tuple. This exposes a direct setup-time tradeoff. A deployment may spend additional one-time work to check every weight-one message and thereby use a more aggressive tuple, such as a smaller k or a larger target distance. Alternatively, it may omit this scan and select a more conservative tuple whose unfiltered bound, including $T(1)$, already meets the same setup budget.

For fast parameter screening, let $S_1 := T(1)$. Exact IOWE sums over a selected set of small intermediate weights give a lower bound $L_1 \leq S_1$, while the exact first-accumulator prefix together with the $U_\star$ contraction gives an upper bound $S_1 \leq U_1$. In the unfiltered mode, $L_1 > \epsilon_{\text{setup}}$ safely discards the tuple immediately. In the filtered mode, U_1 bounds the expected resampling overhead of the exact setup filter below. The complete range, including every $w \geq 2$, remains certified by Algorithm 1.

5.5 Optional Exact Setup-Time Filtering of Weight-One Messages

For deployments choosing the filtered mode, the union bound supports exact rejection sampling at setup time. Let G be the realized $n \times N$ generator matrix, with rows $g_1, \dots, g_n$. Every weight-one message is ae_i for some $a \in \mathbb{F}_q^\times$, and its codeword is ag_i . Nonzero scaling preserves Hamming weight, so it suffices to compute $\text{wt}(g_i)$ once for every $i \in [n]$. The filter accepts exactly when all these weights exceed $d = \lceil \delta N \rceil - 1$.

This scan does not materialize G . For a unit message, repetition and the first interleaver–weightor produce exactly k weighted events. Their sorted positions and prefix values determine the complete first-accumulator output. Streaming the second interleaver, weightor, and accumulator then computes $\text{wt}(g_i)$ in $O(N)$ field operations and $O(k)$ scratch space. The rows are independent and can be checked in parallel.

Algorithm 2: Exact setup-time weight-one rejection filter.

Input: N, q, k, δ
Output: A sampled RAA code description (P_1, W_1, P_2, W_2)

```

1  $d \leftarrow \lceil \delta N \rceil - 1;$ 
2 while true do
3   Sample independent  $(P_1, W_1, P_2, W_2)$  and fix the resulting code  $G$ ;
4    $\text{restart} \leftarrow \text{false};$ 
5   foreach  $i \in [n]$  do
6     Compute  $\text{wt}(g_i) = \text{wt}(\text{RAA}(e_i))$  by the streaming scan;
7     if  $\text{wt}(g_i) \leq d$  then
8        $\text{restart} \leftarrow \text{true}$  and stop this scan
9   if  $\neg \text{restart}$  then
10    return  $(P_1, W_1, P_2, W_2)$ 

```

Let ϵ_1 be a certified upper bound on $T(1)$ and let $\epsilon_{\geq 2}$ bound $\sum_{w=2}^n T(w)$.

Theorem 7 (Filtered setup bound). *Assume $\epsilon_1 < 1$ and run Algorithm 2. The expected number of sampled code descriptions is at most $(1 - \epsilon_1)^{-1}$, and the returned code satisfies*

$$\Pr[\text{dist}(C) < \delta N \mid \text{accepted}] \leq \frac{\epsilon_{\geq 2}}{1 - \epsilon_1}. \quad (18)$$

Proof. Let $\mathcal{B}_1$ be the event that the sampled code has a bad weight-one message, and let $\mathcal{B}_{\geq 2}$ be the corresponding event for a message of weight at least two. The scan accepts exactly on $\overline{\mathcal{B}_1}$, because every weight-one codeword is a nonzero multiple of one checked row. Therefore $\Pr[\text{accepted}] \geq 1 - \epsilon_1$, which also gives the expected resampling bound. Moreover,

$$\Pr[\mathcal{B}_{\geq 2} \cap \text{accepted}] \leq \Pr[\mathcal{B}_{\geq 2}] \leq \epsilon_{\geq 2}.$$

Dividing by the acceptance probability proves Equation (18). $\square$

The complete scan costs $O(nN) = O(N^2/k)$ field operations and stores the $O(N)$ -size code description, while row checks parallelize directly. At the baseline tuple, the ordinary-floating witnesses give $\log_{10} \epsilon_1 \approx -11.0304$ and $\log_{10} \epsilon_{\geq 2} \approx -21.0817$, so the filtered candidate has $\log_{10} \epsilon_{\text{setup}} \approx -21.0817$. The exact scan of all 2^{18} unit messages completed in 69.43 seconds on the Apple M4 Pro used for our experiments. The scan itself is exact. The displayed probability becomes a verified numerical bound after the directed-rounding replay of the certificate.

5.6 Concrete Parameter Search

Algorithm 1 evaluates one fixed tuple. The setup-time RAA-PARAMSELECT algorithm enumerates candidate (k, δ) values and uses RAA-CERTIFY as a feasibility oracle. It separates the returned bound into ϵ_1 and $\epsilon_{\geq 2}$. For a deployment policy $b_{\text{filter}} \in \{0, 1\}$, its setup bound is

$$\epsilon_{\text{mode}} = \begin{cases} \epsilon_1 + \epsilon_{\geq 2}, & b_{\text{filter}} = 0, \\ \epsilon_{\geq 2}/(1 - \epsilon_1), & b_{\text{filter}} = 1. \end{cases} \quad (19)$$

The first line certifies a code sampled once without an exact scan. The second invokes Algorithm 2 and follows from Theorem 7. The selector retains a tuple only when $\epsilon_{\text{mode}} \leq \epsilon_{\text{setup}}$, and then chooses the IOPP query count ℓ and extension degree e according to the independent per-proof soundness bound

$$\epsilon_{\text{total}} \leq (1 - \delta/3)^\ell + N/q^e. \quad (20)$$

It returns the accepted tuple minimizing the specified deployment-cost function, which may account separately for the one-time $O(N^2/k)$ setup scan and the per-proof cost. Thus the result is optimal over the supplied candidate grids. Under the same failure budget, the filtered mode can therefore admit higher-rate tuples that the unfiltered mode rejects because of their weight-one contribution. The setup event and per-proof soundness remain separate budgets: the former is checked once when sampling the code, while the latter is enforced for every proof. Full selector pseudocode appears in Appendix E.3. Correctness follows from the interval enclosures. Interval choices and floating-point optimization affect running time and tightness, and failed verification causes refinement or FAIL.

6 Application: Blind PCS and End-to-End Blind SNARK

The Generalized RAA code (Section 4) is the core algorithmic ingredient that unlocks the first blind PCS with strictly linear $O(N)$ prover work at constant FHE multiplicative depth. Following the standard Ligero-style modular blueprint, our blind PCS is built on top of an *FHE-friendly IOPP* (Interactive Oracle Proof of Proximity) for the RAA code; composing this PCS with a depth-controlled Spartan PIOP for R1CS then yields the end-to-end blind SNARK we implement and evaluate. Our central technical contributions at this layer are (i) the IOPP itself and (ii) the depth-controlled sumcheck folding that preserves $O(1)$ FHE depth across the PIOP at an $O(\log N)$ -factor cost in `mult_plain` work.

Section organization. Section 6.1 constructs the FHE-friendly IOPP for Generalized RAA codes; this is the core. Section 6.2 composes the IOPP with a depth-controlled Spartan PIOP to prove R1CS; its sumcheck is the source of the $O(\log N)$ factor in total work. Section 6.3 uses the IOPP as a black box to build the blind PCS for multilinear polynomial evaluation, the strictly-linear, constant-depth component. Section 6.4 presents an FHE-aware SIMD ciphertext repacking optimization. Section 6.5 compares the assembled SNARK against prior work asymptotically; concrete prover times are in Section 7.

6.1 FHE-Friendly IOPP for the Generalized RAA Code

The IOPP construction follows the Ligero proximity-test framework [3], instantiated with the Generalized RAA code as the underlying linear code and adapted to operate on FHE-encrypted matrix entries. We first state the interface that the PIOP and PCS will consume, then give the construction and analyze its FHE cost.

Interface. An IOPP for a linear code $C \subseteq \mathbb{F}_q^{\hat{n}}$ is a protocol $\langle \mathcal{P}, \mathcal{V} \rangle$ with three operations:

- **Commit**($\{\llbracket m_i \rrbracket\}_{i=1}^r$) $\rightarrow$ **com**: the prover encodes each input row $\llbracket m_i \rrbracket$ with C and Merkle-commits the resulting encoded matrix.
- **LinComb**(γ) $\rightarrow$ $\llbracket u \rrbracket$: given a verifier-chosen vector $\gamma \in \mathbb{F}_{q^e}^r$ and a uniformly sampled query set $I \subseteq [\hat{n}]$ of size ℓ , the prover returns the encrypted random linear combination $\llbracket u \rrbracket = \sum_i \gamma_i \llbracket m_i \rrbracket$ together with openings of the committed encoded matrix at the columns in I .
- **Verify**: the verifier, holding **sk**, verifies the Merkle openings, decrypts the opened columns and the response $\llbracket u \rrbracket$, then checks in plaintext (a) that each opened column is consistent with u at the corresponding output index, and (b) that u is δ -close to a codeword by re-encoding it and comparing with the opened columns at the test positions.

The soundness error of any single **LinComb** query is $(1 - \delta/3)^\ell + N/q^e$, where δ is the relative minimum distance of C , ℓ is the number of column openings, and e is the extension-field degree used by the Schwartz–Zippel check on the linear combination.

Construction (with SIMD packing). Algorithm 3 instantiates this interface with the Generalized RAA code. The input vector is arranged as an $r \times m$ matrix ($r = 2^s$, $rm = n$, with $r \approx m \approx \sqrt{n}$); each row is RAA-encoded to a length- $\hat{n} = m/\rho$ codeword, and the encoded matrix is Merkle-hashed column by column. To exploit BGV/BFV’s $B = 2^{13}\text{--}2^{14}$ SIMD slots, we pack B consecutive matrix rows into the same column-aligned ciphertext: because RAA encoding is element-wise within each row (permutations and per-row weights are public; accumulator scans run along columns), the packing introduces no cross-row dependencies, and one ciphertext operation encodes B rows in parallel.

Cost and soundness. Commit costs $\frac{2n}{B\rho}$ pt-ct multiplications and the same order of ct-ct additions at depth 1 (the per-row weight W_r is the only non-linear step); **LinComb** is a depth-0 pt-ct linear combination, and **Verify** runs in plaintext after the verifier decrypts. The encrypted-data portion of the proof is ℓ column openings plus $\llbracket u \rrbracket$, i.e. $O(\lambda\sqrt{n})$ encrypted field elements before the optimizations of Section 6.4. Conditional on the selected code having relative distance δ , the soundness of a single **LinComb** query is $(1 - \delta/3)^\ell + N/q^e$. For λ -bit security one sets $\ell = \lceil \lambda / \log_2(3/(3 - \delta)) \rceil$ and $e = \lceil (\lambda + \log_2 n) / \log_2 q \rceil$; Table 2 gives the resulting (ℓ^*, e^*) at standard target levels for $\delta = 0.30$.

The IOPP parameters in the table control per-proof soundness. The one-time code-sampling risk is a separate budget: before deployment, Algorithm 1 must certify the selected (N, q, k, δ) tuple. If the rate-1/4 candidate does not meet the chosen setup budget, Algorithm 4 selects a lower rate or returns **FAIL**; this choice does not change the IOPP query count at a fixed δ .

Algorithm 3: FHE-Friendly IOPP for Generalized RAA (SIMD-packed)

Input: SIMD-packed rows $\{\llbracket \mathbf{m}_{b,j} \rrbracket\}_{b \in [\lceil r/B \rceil], j \in [m]}$ where $\mathbf{m}_{b,j} \in \mathbb{F}_q^B$ holds the B rows of column j in block b .

```

// Commit
1  $\{\llbracket \mathbf{c}_{b,j} \rrbracket\}_{j \in [\hat{n}]} \leftarrow E_C(\llbracket \mathbf{m}_{b,1} \rrbracket, \dots, \llbracket \mathbf{m}_{b,m} \rrbracket)$  for each block  $b$ ;          /* RAA
   encode in parallel across the  $B$  SIMD slots; depth 1,  $O(m)$ 
   pt-ct ops per block */
2  $\text{com} \leftarrow \text{MT.Commit}(\{\llbracket \mathbf{c}_{b,j} \rrbracket\})$ ; /* Merkle by column; one leaf per
   packed ciphertext */
3 ;

// LinComb (verifier  $\gamma \in \mathbb{F}_{q^e}^r$ )
4  $\mathcal{V}$  samples  $I$  uniformly among the size- $\ell$  subsets of  $[\hat{n}]$ ;
5  $\llbracket \mathbf{u}_b \rrbracket \leftarrow \sum_{i \in \text{block } b} \gamma_i \cdot \llbracket \mathbf{m}_{b,i} \rrbracket$ ;  $\mathcal{P}$  opens columns  $j \in I$  with Merkle proofs;

// Verify
6  $\mathcal{V}$  verifies the Merkle openings, decrypts  $\llbracket \mathbf{u} \rrbracket$  and the opened columns,
   and checks  $E_C(\mathbf{u})_j \stackrel{?}{=} \sum_b \gamma_{(b)} \mathbf{c}_{b,j}$  slot-wise for  $j \in I$  in plaintext;
```

Table 2. IOPP parameters. For the rate-1/4 code at $\delta = 0.30$, $q = 2^{16} + 1$, the minimum IOPP query count ℓ^* and extension-field degree e^* achieving runtime soundness $\epsilon_{\text{total}} \leq 2^{-\lambda}$ at each security level (per-proof soundness is independent of the code length; the one-time setup risk is treated separately in Section 5.6). Only (ℓ, e) scale with λ .

λ	ℓ^*	e^*	$\log_2(1 - \delta/3)^{\ell^*}$	$\log_2 N/q^{e^*}$	$\log_2 \epsilon_{\text{total}}$
80	527	7	-80.10	-90.00	-80.10
100	658	8	-100.02	-106.00	-100.02
128	843	10	-128.14	-138.00	-128.14

6.2 Composing the PCS with a Depth-Controlled Spartan PIOP

For a public R1CS instance $(A, B, C) \in \mathbb{F}_q^{n \times n}$ and encrypted witness $\llbracket z \rrbracket \in \mathbb{F}_q^n$, we discharge $Az \circ Bz = Cz$ via Setty’s Spartan PIOP [26]: two sumchecks reduce R1CS to one multilinear evaluation $\tilde{z}(r_y) = v_z$, discharged by our blind PCS (Section 6.3, with $\tilde{f} := \tilde{z}$). We choose Spartan over a direct Hadamard-product PIOP because the latter would require n ct-ct multiplications (one per Hadamard slot), whereas Spartan’s degree-3 sumcheck spreads them geometrically across $\log n$ rounds for an aggregate $O(n)$ ct-ct cost.

Depth-controlled sumcheck folding. Each sumcheck round picks a challenge r_i and “folds” the working ciphertext array by setting $\llbracket a_j \rrbracket \leftarrow \llbracket a_{2j} \rrbracket + r_i(\llbracket a_{2j+1} \rrbracket - \llbracket a_{2j} \rrbracket)$. Done in place across $\log n$ rounds, this is $O(n)$ mult_plain work but $\log n$ FHE multiplicative depth, since each round’s mult_plain chains on the previous round’s already-folded ciphertexts, incompatible with our $O(1)$ -depth target.

To preserve $O(1)$ depth, we instead recompute, at each round i , the partial fold $\llbracket a_j^{(i)} \rrbracket := \sum_{b \in \{0,1\}^i} \tilde{\text{eq}}((r_1, \dots, r_i), b) \cdot \llbracket a_{(j \parallel b)} \rrbracket$ from the *original* ciphertext array. Each `mult_plain` now acts on a depth-0 ciphertext, contributing only one level regardless of i ; per-round work rises to $O(n)$ (totalling $O(n \log n)$ over $\log n$ rounds) in exchange. This is the explicit depth-versus-work trade: a $\log n$ factor in cheaper `mult_plain` for constant FHE depth. Per-step costs and the full protocol description appear in Appendix F.1; soundness is the standard Spartan analysis plus the PCS soundness, $\epsilon_{\text{PIOP}} \leq (1 - \delta/3)^\ell + 7N/q^e$.

End-to-end FHE cost. Combining matrix–vector products ($O(n)$ pt-ct, depth 0), the two sumchecks ($O(n)$ ct-ct plus $O(n \log n)$ `mult_plain`, depth ≤ 2 each), and the downstream PCS evaluation ($O(n)$ pt-ct at depth 1), the end-to-end SNARK prover performs $O(n)$ ct-ct multiplications and $O(n \log n)$ total FHE operations at total FHE depth $O(1)$.

6.3 Blind PCS for Multilinear Polynomial Evaluation

For a μ -variate multilinear $\tilde{f}$ with $N = 2^\mu$ encrypted coefficients arranged as an $r \times m$ matrix ($r = 2^s$, $m = 2^{\mu-s}$, $s = \lceil \mu/2 \rceil$), we obtain a blind PCS by composing the IOPP with the standard Ligero-style tensor decomposition $\tilde{f}(z) = \langle q_2, \sum_i q_{1,i} \tilde{f}_i \rangle$ where $q_1 = \bigotimes_{i < s} (1 - z_i, z_i) \in \mathbb{F}_q^r$ and $q_2 = \bigotimes_{i \geq s} (1 - z_i, z_i) \in \mathbb{F}_q^m$. Concretely the prover (i) runs `IOPP.COMMIT`($\{\llbracket \tilde{f}_i \rrbracket\}$) at commit time, then for evaluation at z , (ii) runs `IOPP.LINCOMB`(γ) on a verifier-random γ for proximity testing and (iii) runs `IOPP.LINCOMB`(q_1) on the tensor-structured q_1 to obtain $\llbracket f'' \rrbracket = \sum_i q_{1,i} \llbracket \tilde{f}_i \rrbracket$, which the verifier decrypts and then outputs $y = \langle q_2, f'' \rangle$ in plaintext. Total cost: $O(N)$ FHE operations at depth 1, strictly linear, dominated by the IOPP commit phase; soundness $\epsilon_{\text{PCS}} \leq (1 - \delta/3)^\ell + N/q^e$ (cf. Equation (20)). The security proof (completeness, binding, knowledge soundness) follows the Ligero/Brakedown framework and is deferred to Appendix B; the complete protocol is given as Algorithm 6 in Appendix F.3.

6.4 IOPP-Layer Optimization: Ciphertext Repacking

The base IOPP of Algorithm 3 produces a proof of ℓ Merkle-opened columns plus the encrypted linear-combination response $\llbracket u \rrbracket$. We compress the response with an FHE-aware repacking optimization that operates purely at the IOPP layer and therefore benefits the PIOP and the PCS uniformly.

Ciphertext repacking (Algorithm 5, Appendix F.2). The `LinComb` response is the length- $\hat{n}$ vector $\llbracket u \rrbracket^* = \sum_i \gamma_i \mathbf{f}_i$, but because data is SIMD-packed by column, each scalar u_j^* initially lives alone in its own column ciphertext (using 1 of B slots), so $\llbracket u \rrbracket^*$ is spread over $\hat{n}$ ciphertexts with the other $B - 1$ slots wasted. A two-step *rotate-and-sum* / *mask-and-add* procedure (Algorithm 5) collapses each column to its scalar and then gathers all $\hat{n}$ scalars into $\lceil \hat{n}/B \rceil$ packed ciphertexts, a single ciphertext when $\hat{n} \leq B$ (true for all $N \leq 2^{25}$ at standard SIMD batch sizes) and an $\hat{n}$ -fold reduction in transmitted ciphertexts. It costs $\hat{n} \log_2 B$ rotations plus $O(\hat{n})$ pt-ct mults at zero additional multiplicative depth.

Table 3. Asymptotic comparison of blind SNARK constructions (n = number of gates, N = BFV/BGV SIMD slot count, d = circuit depth). “Ct-ct mults” counts the dominant FHE operation; “Total work” counts all FHE operations including the cheaper `mult_plain`. Measured prover-time / proof-size numbers appear in Section 7.

Scheme	Ct-ct mults	Total work	FHE Depth	Proof Size	IOP / IOPP Layer
PHALANX [30]	$O(kn^{1+1/k})$	$O(kn^{1+1/k})$	$k + O(1)$	$O(k\sqrt{n} + kN \log n)$	RS-IOPP via 2D-NTT
LAMINATE [23]	$O(n)$	$O(n \log n)$	$O(1)$	$O(d \log n)$	GKR + sumcheck (no IOPP)
Ours	$O(n)$	$O(n \log n)$	$O(1)$	$O(\lambda\sqrt{n})$	RAA-IOPP + Spartan PIOP

6.5 End-to-End Blind SNARK and Comparison

Combining the FHE PCS (Section 6.3) with the depth-controlled Spartan PIOP (Section 6.2), both built on the IOPP of Section 6.1 with the optimizations of Section 6.4, yields the end-to-end blind SNARK. Table 3 compares it asymptotically against prior work.

Our construction achieves linear $O(n)$ ct-ct multiplications at constant FHE multiplicative depth, with an $O(\log n)$ overhead in total work coming from the depth-controlled Spartan sumcheck (Section 6.2). The PCS layer alone (Section 6.3) is strictly $O(n)$ at $O(1)$ depth, the first blind PCS to attain this Pareto point. At the SNARK level, our asymptotic profile matches LAMINATE ($O(n \log n)$ total work at $O(1)$ depth), while supporting arbitrary R1CS rather than layered uniform circuits, and improving over PHALANX’s super-linear $O(n^{1+1/k})$ work at constant depth. Our $O(\lambda\sqrt{n})$ proof is larger than LAMINATE’s folding-based proof, the trade-off we accept in exchange for arbitrary-R1CS support at constant FHE depth.

Why the SNARK-level $\log n$ factor. The PCS layer of our construction sidesteps any sum-check via Ligero-style proximity testing, and the Generalized RAA code provides the linear-time, constant-depth encoding that makes proximity testing efficient under FHE. The remaining $\log n$ factor in the SNARK enters *only* at the PIOP layer, in Spartan’s sumcheck. In the FHE setting, a standard in-place sumcheck fold takes $O(n)$ work but contributes $\log n$ multiplicative depth (each round’s `mult_plain` chains on the previous round’s already-folded ciphertexts); preserving $O(1)$ depth via the recompute-from-scratch fold of Section 6.2 costs an extra $\log n$ factor in `mult_plain` work. Whether a depth-controlled sumcheck can be made *strictly linear* under FHE (which would yield a fully linear blind SNARK on top of our PCS) is, to our knowledge, open.

7 Performance Evaluation

We report end-to-end prover time and proof size for the full blind SNARK pipeline: PIOP (Spartan) composed with the RAA-based PCS, all stages run over FHE ciphertexts. All measurements are taken on the same machine and BFV parameter set, namely an Apple M4 Pro (24 GB RAM, single-thread) with Microsoft SEAL using BFV: ring degree $N_{\text{ring}} = 2^{14}$, plaintext modulus

$q = 2^{16} + 1$ (Fermat prime), 3 RNS limbs ($[30, 50, 50]$ bits). Per-proof soundness target is 128-bit with $\delta = 0.30$ and code rate $1/k = 1/4$, giving IOPP query count $\ell^* = 843$ from the parameter-selection procedure of Section 5.6.

Scope of comparison. Neither PHALANX [30] nor LAMINATE [23] has released source code at the time of writing, and by their own description both report *estimated* prover times rather than end-to-end measurements: PHALANX’s Table 3 is labelled “estimated performance” [30, §6], and LAMINATE provides “an estimation of the runtime . . . using detailed operation counts and microbenchmarks” [23, §1.5]. The concurrent LASAGNE [31] likewise *projects* its scaling table by substituting microbenchmarked per-operation constants into operation counts [31, §7], measuring only a single application end-to-end. Our comparison therefore takes the form: (i) we present *measured* single-thread prover times from the `e2e-full` benchmark binary in our codebase (to our knowledge the first end-to-end *measured* blind-SNARK implementation at this scale, provided as supplementary material); (ii) we plot prior-work prover times as *paper-reported* (projected) anchors and extrapolate via each scheme’s own asymptotic formula: PHALANX [30, Table 3] reports 2.27 hr at $N = 2^{20}$ on Intel Xeon Silver 4310 (0.68 hr excluding witness packing, the apples-to-apples figure since our prover also assumes a pre-packed witness), LAMINATE [23, Table 20] reports (single-instance, $N = 2^{20}$, Google Cloud N4) 13.9 hr (base) / 43.9 hr (fold) / 1.1 hr (laned).² Our hardware is an Apple M4 Pro; an Apple Silicon core is typically 1.5–2× faster than these baselines on single-thread SEAL workloads, so the headline factors we report below should be discounted accordingly.

What the three schemes prove, and why direct comparison is subtle. PHALANX proves n -wire R1CS in the regime where the payload computation uses a *single* SIMD slot per ciphertext (Phalanx’s prover-internal SIMD parallelism accelerates its Reed–Solomon encoding, but the R1CS itself is non-batched, see [23]). LAMINATE proves *layered* arithmetic circuits via a GKR-based BhIOP. Its fastest variant, LAMINATE_{laned}, additionally requires the payload to be *laned*: B disjoint instances of the same per-instance circuit, packed one per SIMD slot, with no rotation gates (so the batch is capped at the slot count $B = 2^{14}$, with “free batching” up to that cap); its LAMINATE_{fold} variant lifts the laned restriction to wide (rotation-bearing) layered circuits, but at substantially higher prover cost. Either way the payload must be a layered circuit. Our scheme proves *arbitrary* R1CS via Spartan and an RAA-IOPP, with no layered structure required; SIMD slots carry distinct wire values of one R1CS instance, and the prover scales linearly in the total wire count.

² We use LAMINATE’s May 2026 revision [23], which corrects a soundness issue in its transcript-packing compiler (a randomized well-formedness test adding λ ciphertexts); the fix increases the folded proof size from the 0.27 MB reported in earlier versions (and still cited by concurrent work) to 13.45 MB, and the figures above are the corrected Table 20 numbers. The *laned* variant attains its 1.1 hr only on *laned* payload circuits without rotation gates; the *fold* variant is the one supporting the wide/arbitrary circuits comparable to our R1CS.

7.1 End-to-End Prover Time: Total vs. Paper-Reported SOTA

Figure 2 compares our measured total prover time against the paper-reported numbers for PHALANX and the two principal LAMINATE variants, extrapolated to common N values via each paper’s own asymptotic formula. Concretely we measure 1.46 s, 6.11 s, and 25.5 s at $N = 2^{18}, 2^{20}, 2^{22}$ respectively; the per-doubling-of- N growth factor is $\sim 4.2\times$, essentially linear in N : the $O(\log N)$ overhead from the depth-controlled sumcheck (Section 6) is barely visible in this range, since $\log N$ only varies from 18 to 22.

Versus Phalanx. At $N = 2^{20}$ R1CS wires, PHALANX’s reported 2.27 hr includes witness packing; its core prover excluding packing is 0.68 hr (2,448 s), and since our measurement likewise assumes a pre-packed witness, the apples-to-apples comparison is 0.68 hr versus our 6.11 s, a $\sim 400\times$ speedup (modestly overstated by the hardware mismatch). The gap is structural: PHALANX’s Reed–Solomon encoding uses the low-depth 2D-NTT decomposition introduced by HELIOPO-LIS [4] and costs $O(N^{1+1/k})$, whereas our RAA code encodes in linear time at constant FHE depth.

Versus Laminat (*per-instance cost depends on batch fill*). LAMINATE comes in three variants [23, Table 20], of which only LAMINATE_{fold} supports the wide (cross-slot, rotation-bearing) circuits comparable to arbitrary R1CS; LAMINATE_{laned} is restricted to *laned* payload circuits with no rotation gates, and LAMINATE_{base} trades a 2,604 MB proof for speed. Since LAMINATE SIMD-packs up to $B = 2^{14}$ instances per proof, its per-instance cost spans a band, from the single-instance figure down to the full-batch amortized figure (total runtime/ B); our cost is a flat 6.11 s at $n = 2^{20}$ regardless of batch, since we SIMD-pack one instance’s distinct wires. Taking LAMINATE_{fold}, the apples-to-apples variant, its single-instance cost is 43.9 hr versus our 6.11 s, a $\sim 26,000\times$ per-instance speedup; even at a *full* 2^{14} batch of identically-structured circuits it amortizes only to ≈ 9.7 s/instance, still slower than our batch-independent 6.11 s. The one configuration that proves faster than us is the restricted LAMINATE_{laned} at near-full batch ($15.8 \text{ hr}/2^{14} \approx 3.5 \text{ s/instance}$), which needs 2^{14} identical rotation-free *laned* circuits, exactly the batchable-uniform structure our scheme does not require; single-instance, LAMINATE_{laned} is 1.1 hr, i.e. we are $\sim 650\times$ faster. On proof size, at $n = 2^{20}$ our ~ 32 MB is $\sim 81\times$ smaller than LAMINATE_{base} (2,604 MB) and only $\sim 2.4\times$ larger than LAMINATE_{fold}/LAMINATE_{laned} (~ 13.4 MB), a far narrower size gap than earlier LAMINATE versions reported (see the footnote above).

7.2 Per-Phase Breakdown of Our Prover

Figure 3 shows where the time goes inside our prover. Across all three sizes, the Spartan PIOP dominates ($\geq 94\%$ of total prover time); the entire PCS pipeline sums to under 0.9 s even at $N = 2^{22}$: RAA encoding (0.02–0.45 s), Merkle commit (0.02–0.34 s), LinComb (8–70 ms), column openings (6–10 ms), and

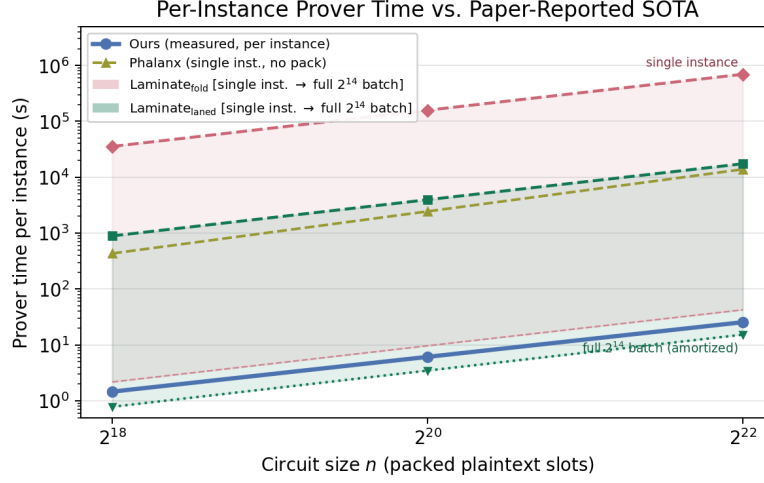


Fig. 2. Per-instance prover time (log y -axis). Each LAMINATE variant from the corrected Table 20 [23, Table 20] is drawn as a *band* spanning its per-instance cost from a single instance (upper edge) down to a full $B = 2^{14}$ batch amortized (lower edge): LAMINATE_{fold} (red) supports wide/arbitrary circuits, while LAMINATE_{laned} (teal) is the fastest but restricted to laned rotation-free circuits. PHALANX [30, Table 3] cannot SIMD-batch its R1CS, so it is a single-instance line (shown excluding witness packing, apples-to-apples with our pre-packed-witness setting). Our per-instance cost (solid blue) is fixed regardless of batch, since we SIMD-pack one instance’s distinct wires; it sits orders below every single-instance edge (about 26,000 $\times$ below LAMINATE_{fold} at $n = 2^{20}$), below LAMINATE_{fold}’s full-batch edge, and is undercut only by LAMINATE_{laned} at a full 2^{14} batch of identical rotation-free circuits. Prior-work numbers are extrapolated to other N via each paper’s stated asymptotic scaling.

repacking of u^* (33 ms). The repack phase is essentially N -independent because the LinComb response u^* fits in a single SIMD-packed ciphertext (Section 6.4); repacking 1 ciphertext costs $\log_2(B/2) + 1$ rotations regardless of N .

What the breakdown says about the asymptotic claims. The breakdown empirically corroborates the asymptotic split (Section 6): the PCS phases scale at most linearly with essentially flat per-wire cost, while the only super-linear component is the depth-controlled Spartan sumcheck, whose $O(\log N)$ factor in `mult_plain` work is the sole source of the end-to-end $O(N \log N)$ term (the ct-ct multiplications remain $O(N)$).

Counterfactual code cost. The subsecond PCS contribution in Figure 3 depends on the linear-time, depth-1 RAA encoding. Replacing it with the low-depth Reed–Solomon encoding used by Phalanx would require $O(kN^{1+1/k})$ homomorphic operations at depth $k + 1$. For every fixed k , this exceeds the $O(N \log N)$ Spartan cost asymptotically and therefore moves the bottleneck back to the PCS.

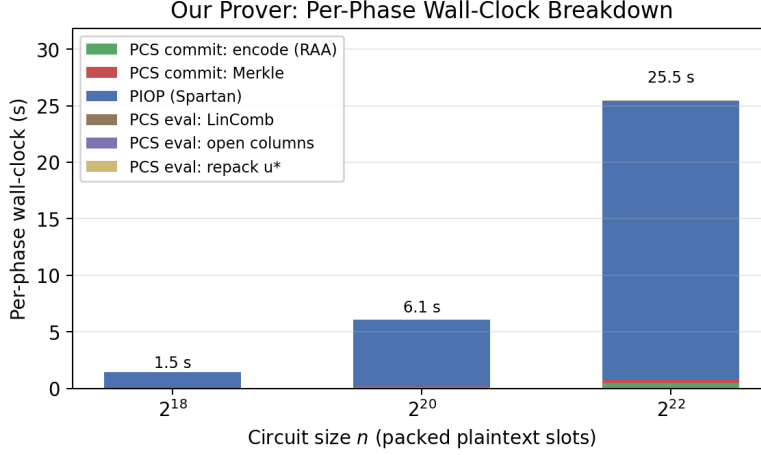


Fig. 3. Per-phase prover-time breakdown of our prover (linear y -axis). Stacked bars are our **e2e-full** measurements on Apple M4 Pro at $N \in \{2^{18}, 2^{20}, 2^{22}\}$; totals are annotated above each bar. The Spartan PIOP dominates uniformly across the range; all PCS phases together are under 5% of total prover time.

Brakedown retains $O(N)$ encoding operations, but its $O(\log N)$ multiplicative depth enlarges the ciphertext modulus, and hence the cost of each homomorphic operation, by the same asymptotic factor. Its effective ring-arithmetic cost is therefore $O(N \log N)$, of the same order as the PIOP rather than the negligible PCS fraction measured here.

7.3 Proof Size

We do not include a proof-size figure because the comparison does not have a single clean axis of advantage. Concretely, our proof at $N = 2^{22}$, $\lambda = 128$, $\ell^* = 843$ is ~ 105 MB (dominated by the 843 Merkle-opened codeword columns plus one packed-ciphertext LinComb response after SIMD repacking). At $N = 2^{20}$ it is ~ 32 MB. For context, PHALANX reports 58 MB at $N = 2^{20}$ and an $O(\sqrt{N})$ growth that reaches ~ 116 MB at $N = 2^{22}$ [30, Table 3], essentially the same order as ours. LAMINATE [23, Table 20] reports, at $n = 2^{20}$, 2,604 MB (base), 13.45 MB (fold), and 13.33 MB (laned); the two compact variants land within a factor of ~ 2.4 of our proof (corrected May 2026 figures; see the comparison-scope footnote above).

On the time-size frontier, then, our proof is $\sim 81\times$ smaller than LAMINATE_{base} (2,604 MB) at $n = 2^{20}$ and only $\sim 2.4\times$ larger than LAMINATE_{fold}/LAMINATE_{laned} (~ 13.4 MB), while our single-instance prover is orders of magnitude faster (Section above). We do not claim a proof-size win; the chosen operating point is the trade-off we accept in exchange for arbitrary-R1CS support: a moderate proof and fast prover (linear in ct-ct multiplications, near-linear in total FHE work; see Section 6), with no circuit-shape constraint.

7.4 Multiplicative Depth and Per-Operation Cost

Holding the circuit at 2^{22} gates and $\delta = 0.30$ fixed, the only quantities that vary with the per-proof soundness target λ are the IOPP query count ℓ^* and extension-field degree e^* (Table 2, Section 6.1). The one-time code-sampling risk is independent of λ and must be checked for the chosen per-row code length by Algorithm 1.

Depth-cost. Our scheme uses 3 RNS limbs (1 multiplicative level for the per-row weightor at depth $0.5 + 1$ level for the Spartan PIOP’s ct-ct multiplication $+ 1$ base, totalling $\log Q \approx 130$ bits). Brakedown’s expander codes require $O(\log N)$ levels, negating their linear encoding advantage under FHE; the Reed–Solomon 2D-NTT of HELIOPOLIS [4], as used by Phalanx, requires $k + 1$ levels at k -way decomposition. Per-ciphertext-op cost scales as $\tilde{O}(M \log Q) = \tilde{O}(M \cdot d)$ (Section 1), so a $7\times$ depth gap (Brakedown at $N = 2^{30}$) translates to a $\sim 7\times$ per-op cost penalty over our $1\times$ baseline.

8 Conclusion and Future Work

We established linear minimum distance with failure probability $o(1/N)$ for Generalized RAA codes over any fixed non-binary field, via the Pairwise-Product IOWE Bound (Lemma 2), together with an efficient concrete failure-probability certifier and setup-time parameter-search procedure (Algorithm 1). Combined with a Ligero-style proximity test this gives the first blind PCS with strictly linear $O(N)$ prover work at $O(1)$ FHE depth; composing it with a depth-controlled Spartan PIOP for R1CS yields, to our knowledge, the first end-to-end *measured* blind SNARK implementation at this scale, with $O(N)$ ct-ct multiplications and $O(N \log N)$ total FHE work at constant depth. Empirically (Section 7) the prover scales near-linearly in N ($\sim 4.2\times$ per doubling), in clear contrast to PHALANX’s super-linear $O(kN^{1+1/k})$ growth.

Limitations and future work. At $N = 2^{20}$, our ~ 32 MB proof is smaller than PHALANX’s 58 MB proof and within $\sim 2.4\times$ of LAMINATE’s compact ~ 13.4 MB variants. At $N = 2^{22}$, our proof is ~ 105 MB, comparable to PHALANX’s ~ 116 MB proof (Section 7). Reducing this size while retaining constant FHE depth is a natural refinement. The concrete distance bound covers all intermediate and message weights without empirical tail extrapolation. Tightening its analytic upper bounds, completing portable directed-rounding verification of the full-range calculation, and making the depth-controlled sumcheck strictly linear under FHE remain useful directions.

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A Numerical Illustration

This section illustrates two features of the analysis. First, it shows how the low-output probability $D(h)$ and the normalized accumulator IOWE $\tilde{A}(w, h)$ vary with their input weights. These plots explain why the asymptotic proof uses different inequalities in its small- and large-weight branches. Second, it plots the per-weight union-bound contribution $T(w)$ from Equation (6). The observed shapes guide the choice of weight intervals, but they are not assumptions in the proof.

Exponential decay of $D(h)$ across (N, δ)

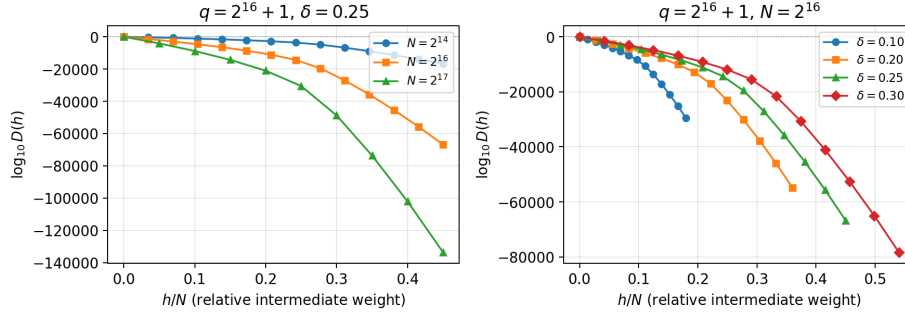


Fig. 4. Low-output probability $D(h)$ at $q = 2^{16} + 1$. *Left:* fixed $\delta = 0.25$ and varying $N \in \{2^{14}, 2^{16}, 2^{17}\}$. *Right:* fixed $N = 2^{16}$ and varying $\delta \in \{0.10, 0.20, 0.25, 0.30\}$. The horizontal axis is h/N . The near-linear decline on the logarithmic scale illustrates the decay used in the large-weight analysis.

A.1 Why the Proof Splits at $w \approx 0.01 N/k$

$D(h)$ decays exponentially in h . Figure 4 plots $\log_{10} D(h)$, where $D(h)$ is the probability that a weight- h input to the second randomized accumulation produces at most δN nonzero outputs. The curves are close to linear on a logarithmic scale. At fixed $\delta = 0.25$, the slope becomes steeper as N grows. At fixed $N = 2^{16}$, it becomes steeper as δ decreases. This behavior is consistent with the exponential estimate used in the large-weight proof. The formal asymptotic statement is Theorem 3; Theorem 5 gives the explicit finite-parameter bound.

Conditional accumulator output-weight profiles ($N = 2^{14}$, $q = 2^{16} + 1$)

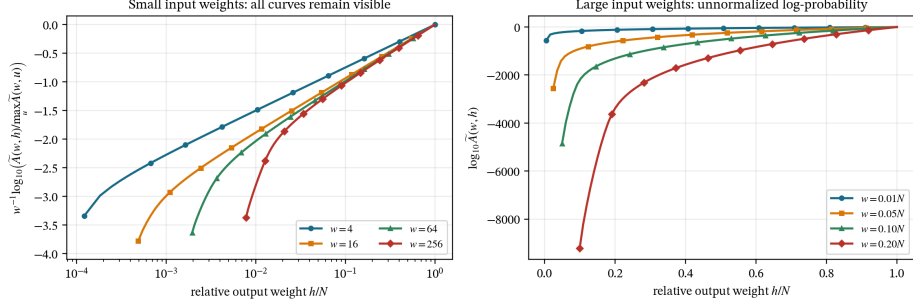


Fig. 5. Conditional accumulator output-weight profiles at $N = 2^{14}$ and $q = 2^{16} + 1$. *Left:* for $w \in \{4, 16, 64, 256\}$, we plot $w^{-1} \log_{10}(\tilde{A}(w, h) / \max_u \tilde{A}(w, u))$ so that every small-weight curve remains visible. Zero denotes the peak of each normalized profile and is not a failure probability. *Right:* for $w \in \{0.01N, 0.05N, 0.10N, 0.20N\}$, we plot the unnormalized quantity $\log_{10} \tilde{A}(w, h)$. The panels illustrate why the proof uses direct combinatorial bounds for small inputs and entropy estimates for inputs of linear weight.

The accumulator distribution has different useful descriptions. Figure 5 plots the conditional probability $\tilde{A}(w, h)$ that a randomized weight- w input produces output weight h . The left panel rescales each small- w curve by its own maximum and by w . This puts all four shapes on a readable common scale, including $w = 4$. The plotted quantity is a relative profile, not a distance-failure probability. In this regime, direct factorial bounds are more useful than Stirling approximations, which motivates Lemma 2. The right panel keeps the unnormalized log-probability for input weights that are constant fractions of N , where entropy estimates are effective.

The figure illustrates the two proof techniques but does not select the cutoff $0.01N/k$. As explained in Section 4.3, 0.01 is a convenient fixed constant that simultaneously supplies slack to the large-weight entropy argument and ensures $N/(N - kw) \leq 100/99$ in the small-weight argument. Other sufficiently small constants would also work after changing the proof constants.

A.2 Empirical Shape of $T(w)$

Figure 6 documents the empirical shape of $T(w) = \binom{n}{w} (q - 1)^w \sum_h P_1(h \mid kw) D(h)$. These profiles motivate the weight partition used in the finite computation, but they are not a correctness assumption: Theorem 6 bounds every message weight even if an untested parameter tuple has a non-monotone profile.

Figure 6 reports $\log_{10} T(w)$ across the deployment regime $q = 2^{16} + 1$, $k = 4$, $\delta \in \{0.10, 0.20, 0.25, 0.30\}$, $N \in \{2^{18}, 2^{20}, 2^{22}\}$. In every slice, $T(w)$ is monotone non-increasing, with the per-step ratio $T(w + 1)/T(w)$ shrinking as w grows. This decay is purely an empirical observation at the deployment scale; we do not claim it as a general theorem.

Monotonic decay of $T(w)$ under Algorithm 1 (per- w adaptive $h_{\max}$, truncated at w_{stop})

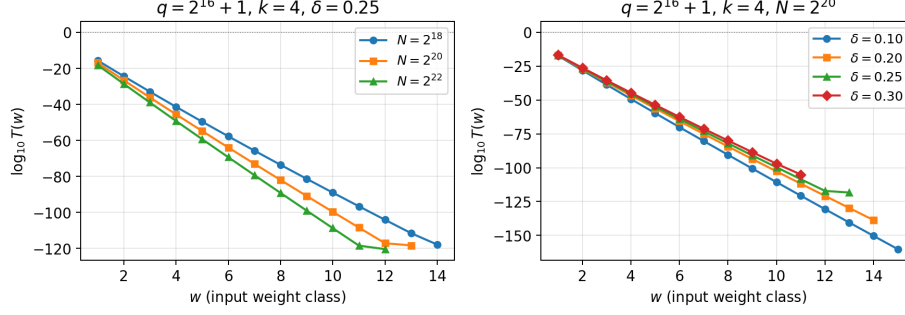


Fig. 6. Monotonic decay of $T(w)$. *Left:* fixed $\delta = 0.25$, varying $N \in \{2^{18}, 2^{20}, 2^{22}\}$. *Right:* fixed $N = 2^{20}$, varying $\delta \in \{0.10, 0.20, 0.25, 0.30\}$. Both slices show strict monotonicity with geometric slope at deployment-scale N . All curves use $k = 4$, $q = 2^{16} + 1$.

B Blind PCS Security Properties

We instantiate the security model from Laminat’s BhIOP framework [23]. Throughout, $\mathcal{E} = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval})$ is an IND-CPA-secure leveled HE scheme, $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ is a hash function modelled as a random oracle, and MT denotes a Merkle commitment scheme built from H in the standard way.

B.1 Security Model: Blind Indexed PCS

Indexed blind language for multilinear PCS. Fix message space $\mathbb{F}_q$, message length n , codeword length N , and let $C : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^N$ denote the RAA encoder. A blind PCS for μ -variate multilinear polynomials is associated with the indexed blind language

$$\mathcal{E}[\mathcal{L}_{\text{PCS}}] = \left\{ ((\text{params}, \text{sk}); (\text{ct}[\tilde{f}], z, y)) : \mathcal{E}.\text{Dec}_{\text{sk}}(\text{ct}[\tilde{f}]) = \tilde{f} \in \mathbb{F}_q^{2^\mu}, \tilde{f}(z) = y \right\}.$$

The blind index is $i' = ((\text{params}, n, N, C), \text{sk})$. Following Laminat’s Definition 2.6, the verifier $\mathcal{V}$ holds sk (designated-verifier setting); the prover $\mathcal{P}$ holds only the evaluation key evk .

Public-coin BhIOP execution. The PCS protocol of Section 6.3 is a 3-message public-coin BhIOP Π : $\mathcal{P}$ ’s commit message $\text{com} = \text{MT.Commit}(\text{ct}[c_1], \dots, \text{ct}[c_r])$, $\mathcal{V}$ ’s challenge $(q_1, \text{idx}_1, \dots, \text{idx}_\ell, \gamma)$ (column-opening indices and proximity-test coefficients), and $\mathcal{P}$ ’s response (encrypted column openings and the linear-combination ciphertext $\text{ct}[f'']$). The Fiat–Shamir compilation derives challenges via H applied to the transcript so far. Throughout we work in the random-oracle model and treat the Merkle/Fiat–Shamir layer as binding except with

probability $\text{negl}(\lambda)$ (the standard preimage and collision events over H), as in the BhIOP framework [23]; we fold these into a single $\text{negl}(\lambda)$ term and do not track hash constants explicitly.

Extractor interface. The knowledge extractor $\mathcal{E}_{\text{kn}}$ is granted three capabilities, matching the BhIOP designated-verifier setting: (E1) black-box access to a malicious prover $\mathcal{P}^*$ (with rewinding); (E2) oracle access to H via the standard ROM simulation (with query $\log Q_H$); (E3) access to the secret-key decryption oracle $\mathcal{E}.\text{Dec}_{\text{sk}}(\cdot)$. (E3) is the FHE analogue of "extractor sees the witness" in plaintext extraction: in the BhIOP model, the verifier's sk is the only thing distinguishing the verifier from a public-coin observer, and granting it to $\mathcal{E}_{\text{kn}}$ is the canonical relaxation [23]. Without (E3) the extractor would have to break IND-CPA security to recover any plaintext, contradicting the FHE assumption.

B.2 Completeness and Binding

Theorem 8 (Completeness). *If $\mathcal{P}$ and $\mathcal{V}$ follow the PCS protocol of Section 6.3 honestly, then $\Pr[\mathcal{V} \text{ accepts}] = 1$.*

Proof. RAA encoding is $\mathbb{F}_q$ -linear, so $\text{Enc}_C(\sum_i \gamma_i \tilde{f}_i) = \sum_i \gamma_i \text{Enc}_C(\tilde{f}_i)$ at every column. Since BGV/BFV homomorphic addition and pt-ct multiplication implement the corresponding plaintext operations exactly (no decryption noise at depth ≤ 1 given the parameter choice), the column-consistency check passes at every opened column. The evaluation output $\langle q_2, f'' \rangle = \tilde{f}(z)$ follows from the multilinear-extension tensor identity.

Theorem 9 (Computational Binding). *Condition on a successful setup (C attains relative minimum distance $\geq \delta$, except with the separately-budgeted one-time risk ϵ_{RAA} of Theorem 2; Section 5.6). Then no PPT adversary opens a commitment com^* to two distinct encrypted polynomials $\text{ct}[\tilde{f}] \neq \text{ct}[\tilde{f}']$ except with probability $\epsilon_{\text{bind}} \leq N/q^e + \text{negl}(\lambda)$.*

Proof. The Merkle commitment binds com^* to a unique tuple of ciphertext rows except with $\text{negl}(\lambda)$ (Appendix B.1); since $\mathcal{E}.\text{Dec}_{\text{sk}}$ is deterministic, these decrypt to a unique tuple of plaintext rows $(c_1, \dots, c_r)$. Under the good-setup conditioning each row lies within the unique-decoding radius $\delta N/2$ of at most one codeword of C , and the column-consistency check forces each opened row to equal that codeword except with probability $\leq N/q^e$ (Schwartz–Zippel over $\mathbb{F}_{q^e}$). The recovered polynomial is thus unique, contradicting $\tilde{f} \neq \tilde{f}'$.

B.3 Knowledge Soundness in the BhIOP Setting

Theorem 10 (Knowledge Soundness). *Condition on a successful setup (C attains relative distance $\geq \delta$, except with the separately-budgeted one-time risk ϵ_{RAA} of Theorem 2; Section 5.6). Then there exists an expected-PPT extractor $\mathcal{E}_{\text{kn}}$ with the interface (E1)–(E3) of Appendix B.1, such that for every PPT prover*

$\mathcal{P}^*$ convincing the *BhIOP* verifier with probability $\geq \varepsilon$, $\mathcal{E}_{\text{kn}}^{\mathcal{P}^*, H, \mathcal{E}.\text{Dec}_{\text{sk}}}$ outputs a multilinear polynomial $\tilde{f} \in \mathbb{F}_q^{2^\mu}$ satisfying $\tilde{f}(z) = y$ with probability at least

$$\varepsilon - \varepsilon_{\text{total}}, \quad \varepsilon_{\text{total}} = \underbrace{(1 - \delta/3)^\ell}_{\text{IOPP soundness}} + \underbrace{N/q^e}_{\text{Schwartz-Zippel}} + \underbrace{\text{negl}(\lambda)}_{\text{ROM/commitment}}.$$

Proof. Executed over plaintexts, the protocol is exactly the Ligerio/Brakedown PCS for the interleaved code C^r ; the only FHE-specific features are that the committed objects are ciphertexts and that the response is SIMD-repacked. We dispatch both and inherit the rest.

(1) *Determinism reduces ciphertext-soundness to plaintext-soundness.* $\mathcal{E}.\text{Dec}_{\text{sk}}$ is a deterministic function, so the ciphertext layer gives a cheating prover no extra freedom: a commitment and its openings decrypt to a single fixed plaintext transcript, and the verifier’s column-consistency and evaluation checks read only those decryptions. Concretely, by Merkle binding (except $\text{negl}(\lambda)$; Appendix B.1) the commitment com^* fixes a unique tuple of ciphertext rows, which $\mathcal{E}_{\text{kn}}$ reads from the query log Q_H via (E1),(E2); decrypting them with (E3) yields a unique tuple of plaintext rows $(c_1, \dots, c_r) \in \mathbb{F}_q^{N \times r}$ —exactly the rows the honest verifier decrypts in-protocol. The execution is therefore identical to the plaintext interleaved-code PCS run on $(c_1, \dots, c_r)$, and soundness over ciphertexts reduces to soundness over those plaintexts.

(2) *SIMD repacking does not affect soundness.* The repacking optimization (Section 6.4, Appendix F.2) changes only the *transmission format* of the *LinComb* response: it deterministically gathers the $\hat{n}$ per-column scalars u_j^* into $\lceil \hat{n}/B \rceil$ packed ciphertexts by public slot rotations and masking, assigning each scalar to a distinct slot (a single ciphertext when $\hat{n} \leq B$), at zero added multiplicative depth. It touches neither the commitment nor the committed rows. This slot map is a public bijection, so the verifier recovers the same u^* by reading the designated slots before running its checks; the checked values—and hence both binding and knowledge soundness—are unchanged.

(3) *Inherited soundness.* $\mathcal{E}_{\text{kn}}$ runs the Ligerio/Brakedown extractor on the decrypted rows, and knowledge soundness is exactly that of the plaintext PCS [3,16]. With ℓ random column openings at proximity parameter $\delta/3$ (within C ’s unique-decoding radius under the good-setup conditioning), that analysis returns $\tilde{f}$ with $\tilde{f}(z) = y$ except with probability $(1 - \delta/3)^\ell + N/q^e$ (testing soundness plus the Schwartz-Zippel evaluation check over $\mathbb{F}_{q^e}$); the random-oracle/commitment layer contributes $\text{negl}(\lambda)$. Hence $\Pr[\text{ExtractOK}] \geq \varepsilon - \varepsilon_{\text{total}}$, and $\mathcal{E}_{\text{kn}}$ runs in expected polynomial time (one transcript, r decryptions, and the plaintext extractor).

C Per-Operation FHE Cost Derivation

The RAA encoding pipeline $c_i = f_i R(P_{\pi_1} W_{\mathbf{r}_1} A)(P_{\pi_2} W_{\mathbf{r}_2} A)$ has the following per-row costs:

- **Repeat & Permute:** data movement only, zero FHE operations, 0 depth.

- **Weight** (W_{r_1}): $\hat{n}$ pt-ct multiplications, 0.5 depth (public weights).
- **Accumulate** (A): $\hat{n} - 1$ ct-ct additions, 0 depth (additions are noise-additive, not multiplicative, in BGV/BFV).
- **Second round**: identical to the first: $\hat{n}$ pt-ct mults, $\hat{n} - 1$ ct-ct adds, 0.5 depth.

Total per row: $2\hat{n}$ pt-ct mults, $2\hat{n} - 2$ ct-ct adds, 1 level. With SIMD packing (B rows per ciphertext): $\frac{2N}{B\rho}$ pt-ct mults, $\frac{2N}{B\rho} - \frac{2r}{B}$ ct-ct adds, 1 level.

D Deferred Proofs from the Distance Analysis

This appendix contains the detailed proofs of the lemmas, theorem, and corollaries used in Section 4.3 (the distance proof of Theorem 2). The statements themselves remain in the main body; only the proofs are deferred here.

D.1 Proof of Theorem 3 (Exponential decay of $D(w)$)

Proof (Proof of Theorem 3). Fix the theorem's threshold η and, within this proof, write $\delta := \eta$ and $D := D_\eta$. We alias $a = \lceil (w - i)/2 \rceil$ and $b = \lfloor (w - i)/2 \rfloor$, so $a - b \in \{0, 1\}$. First we analyze the dominant term in $\tilde{A}(w, h)$ for $h \in [\delta N]$:

$$\begin{aligned} \tilde{A}(w, h) &= \sum_{i=0}^{w-1} \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \cdot \frac{(q-1)^a (q-2)^i}{(q-1)^w} \\ &\leq \sum_{i=0}^{w-1} \frac{\binom{h}{a} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \end{aligned} \quad (21)$$

The parity gap $a - b \leq 1$ changes the logarithm of each adjacent binomial coefficient, and of the denominator when its index is shifted by one, by at most $O(\log N)$. We therefore absorb an $N^{O(1)}$ factor and write $a = \alpha N$, $h = \beta N$, and $i = \theta N$; then $w/N = 2\alpha + \theta + O(1/N)$. Applying the standard entropy bounds for binomial coefficients gives, up to this polynomial factor:

$$\begin{aligned} \binom{h}{a} &\leq e^{H(\alpha/\beta)\beta N}, & \binom{N-h}{b} &\leq e^{H(\alpha/(1-\beta))(1-\beta)N}, \\ \binom{h-a}{i} &\leq e^{H(\theta/(\beta-\alpha))(\beta-\alpha)N}, & \binom{N}{w} &\geq N^{-O(1)} e^{H(2\alpha+\theta)N}. \end{aligned}$$

Substituting into (21):

$$\tilde{A}(w, h) \leq w N^{O(1)} \cdot \max_{0 \leq i \leq w-1} e^{\Phi(\alpha, \beta, \theta)N}$$

where $\Phi = (1 - \beta)H(\frac{\alpha}{1-\beta}) + \beta H(\frac{\alpha}{\beta}) + (\beta - \alpha)H(\frac{\theta}{\beta - \alpha}) - H(2\alpha + \theta)$.

Since $D(w) \leq \delta N \cdot \max_h \tilde{A}(w, h)$, we have:

$$D(w) \leq w N^{O(1)} \max_{\substack{0 \leq i \leq w-1 \\ 1 \leq h \leq \delta N}} e^{\Phi(\alpha, \beta, \theta)N}$$

Maximizing over β . We compute:

$$\frac{\partial \Phi}{\partial \beta} = \log \left(\frac{(1 - \beta - \alpha)\beta}{(1 - \beta)(\beta - \alpha - \theta)} \right)$$

When $\beta \leq \delta < \frac{1}{2}$, the inequality $(1 - \beta - \alpha)\beta > (1 - \beta)(\beta - \alpha - \theta)$ holds, giving $\partial \Phi / \partial \beta > 0$. Hence $\max_{\beta} \Phi = \Phi(\alpha, \delta, \theta)$.

Maximizing over θ . Setting $\partial \Phi / \partial \theta = 0$ yields $\theta^* = \frac{2\alpha(\delta - \alpha)}{1 + \alpha - \delta}$, giving:

$$\begin{aligned} \max_{\theta} \Phi &= \delta H\left(\frac{\alpha}{\delta}\right) + (1 - \delta)H\left(\frac{\alpha}{1 - \delta}\right) - (1 - \delta + \alpha)H\left(\frac{2\alpha}{1 - \delta + \alpha}\right) \\ &\leq \delta H\left(\frac{\alpha}{\delta}\right) + (1 - \delta)H\left(\frac{\alpha}{1 - \delta}\right) - (1 - \delta)H\left(\frac{2\alpha}{1 - \delta}\right) \triangleq \Psi_{\delta}(\alpha) \end{aligned}$$

Monotonicity of Ψ_{δ} .

$$\frac{d\Psi_{\delta}}{d\alpha} = \log \left(1 + \frac{(5\delta - 1)(1 - \delta) - 4\delta\alpha}{(1 - \delta - 2\alpha)^2} \right)$$

When $\delta < 0.2$, $d\Psi_{\delta}/d\alpha < 0$ for all α . Since $2\alpha + \theta \geq 0.01$ and $\theta^* = \frac{2\alpha(\delta - \alpha)}{1 + \alpha - \delta}$, we get $\frac{2\alpha}{1 - \delta + \alpha} \geq 0.01$, hence $\alpha \geq \frac{1 - \delta}{199}$. Thus $\max \Psi_{\delta} = \Psi_{\delta}(\frac{1 - \delta}{199}) < 0$ when $\delta < 0.2$.

Conclusion. Put $\psi_{\eta} := \Psi_{\eta}((1 - \eta)/199) < 0$. For $w \geq 0.01N$,

$$D_{\eta}(w) \leq wN^{O(1)}e^{\psi_{\eta}N}.$$

Because $w \leq N$ and $\psi_{\eta} < 0$, we have $\psi_{\eta}N \leq \psi_{\eta}w$. Moreover, for every fixed $\epsilon > 0$, the condition $w \geq 0.01N$ implies

$$\log(wN^{O(1)}) = O(\log N) \leq \epsilon w$$

for all sufficiently large N . Hence

$$D_{\eta}(w) \leq e^{(\psi_{\eta} + \epsilon)w}.$$

Choose $0 < \epsilon < -\psi_{\eta}$ and set $\Lambda_{\eta} = -\psi_{\eta} - \epsilon > 0$. The explicit constants in Remark 1 use the valid choice $\epsilon = -\psi_{\eta}/2$, hence $\Lambda_{\eta} = -\psi_{\eta}/2$.

D.2 Proofs of Corollaries 1 and 2

Proof (Proof of Corollary 1). Put $\eta_0 := 0.01$. Apply Theorem 3 at thresholds η_0 and δ , obtaining positive constants Λ_{η_0} and Λ_{δ} . Since $kw \geq \eta_0N$, split the true first-accumulator distribution at η_0N :

$$\begin{aligned} \sum_{s=1}^N \tilde{A}(kw, s)D_{\delta}(s) &\leq \sum_{s \leq \eta_0N} \tilde{A}(kw, s) + \sum_{s > \eta_0N} \tilde{A}(kw, s)e^{-\Lambda_{\delta}s} \\ &\leq D_{\eta_0}(kw) + e^{-\eta_0\Lambda_{\delta}N} \\ &\leq e^{-\Lambda_{\eta_0}kw} + e^{-\eta_0\Lambda_{\delta}N} \\ &\leq 2e^{-\mu N}, \end{aligned}$$

where

$$\mu := \eta_0 \min\{\Lambda_{\eta_0}, \Lambda_\delta\} > 0.$$

The complete message-weight prefactor satisfies

$$\binom{N/k}{w} (q-1)^w \leq \sum_{u=0}^{N/k} \binom{N/k}{u} (q-1)^u = q^{N/k}.$$

Therefore the weight- w contribution is at most

$$2 \exp \left[- \left(\mu - \frac{\log q}{k} \right) N \right].$$

Choose $K > \log(q)/\mu$. For any fixed $k > K$, the constant factor 2 can be absorbed by decreasing the positive exponent constant, giving the claimed e^{-CN} bound for all sufficiently large N .

For the explicit constants in Remark 1, the same argument simplifies. Since $\eta_0 > 2\delta_0$, an accumulator whose input has weight $s > \eta_0 N$ cannot have output weight at most $\delta_0 N$: every nonzero output run accounts for at most two nonzero input symbols. Hence $D_{\delta_0}(s) = 0$ on this branch. The inner sum in the proof of Corollary 1 is therefore at most $D_{\eta_0}(kw) \leq e^{-\Lambda_{\eta_0} kw}$, and the large-weight condition reduces to $k > \log(q)/(\eta_0 \Lambda_{\eta_0})$, as recorded in Equation (1). The same support condition makes the $s \geq \eta_0 N$ term in Corollary 2 identically zero.

Proof (Proof of Corollary 2). Put $\eta_0 := 0.01$. Theorem 3 at threshold δ gives

$$\begin{aligned} \sum_{s \geq \eta_0 N} \tilde{A}(kw, s) D_\delta(s) &\leq e^{-\eta_0 \Lambda_\delta N} \sum_{s \geq \eta_0 N} \tilde{A}(kw, s) \\ &\leq e^{-\eta_0 \Lambda_\delta N}. \end{aligned}$$

Using the same uniform prefactor bound $\binom{N/k}{w} (q-1)^w \leq q^{N/k}$, the full contribution is at most

$$\exp \left[- \left(\eta_0 \Lambda_\delta - \frac{\log q}{k} \right) N \right].$$

Choosing $K > \log(q)/(\eta_0 \Lambda_\delta)$ proves the claim.

D.3 Proof of Lemma 2 (Pairwise-Product IOWE Bound)

Proof (Proof of Lemma 2). Alias $a = \lceil \frac{w-i}{2} \rceil$ and $b = \lfloor \frac{w-i}{2} \rfloor$, so $a + b = w - i$ and $a - b \in \{0, 1\}$. Since $a + i \leq (w + i + 1)/2 \leq w$, the nonzero-element factor satisfies $\frac{(q-1)^a (q-2)^i}{(q-1)^w} \leq 1$.

Using $\binom{N}{w} = \frac{N}{w} \binom{N-1}{w-1}$, we get:

$$\frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} \leq \frac{w}{N} \cdot \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N-1}{w-1}}.$$

Also, $w \leq 0.01N$ implies

$$\frac{N}{N-w} \leq \frac{100}{99} < 2.$$

Case $a = b + 1$:

$$\binom{h-1}{a-1} \binom{N-h}{b} = \binom{h-1}{b} \binom{N-h}{b} \leq \frac{h^b N^b}{(b!)^2}.$$

With $\binom{h-a}{i} \leq h^i/i!$ and $\binom{N-1}{w-1} \geq (N-w)^{w-1}/(w-1)!$:

$$\begin{aligned} \frac{\binom{h-1}{a-1} \binom{N-h}{b} \binom{h-a}{i}}{\binom{N}{w}} &\leq \frac{w}{N} \binom{w-1}{a-1, b, i} \left(\frac{h}{N-w} \right)^{a+i-1} \\ &\quad \cdot \left(\frac{N}{N-w} \right)^b \\ &\leq \frac{w \cdot 6^w}{N} \left(\frac{h}{N-w} \right)^{a+i-1}. \end{aligned}$$

Here we used $\binom{w-1}{a-1, b, i} \leq 3^{w-1}$, $\left(\frac{N}{N-w} \right)^b \leq 2^b \leq 2^w$, and $a+i-1 = b+i$.

Case $a = b$:

$$\binom{h-1}{a-1} \binom{N-h}{a} \leq \frac{h^{a-1} N^a}{(a-1)!a!}.$$

Again using $\binom{h-a}{i} \leq h^i/i!$ and $\binom{N-1}{w-1} \geq (N-w)^{w-1}/(w-1)!$:

$$\begin{aligned} \frac{\binom{h-1}{a-1} \binom{N-h}{a} \binom{h-a}{i}}{\binom{N}{w}} &\leq \frac{w}{N} \binom{w-1}{a-1, a, i} \left(\frac{h}{N-w} \right)^{a+i-1} \left(\frac{N}{N-w} \right)^a \\ &\leq \frac{w \cdot 6^w}{N} \left(\frac{h}{N-w} \right)^{a+i-1}. \end{aligned}$$

In both cases, $a+i-1 \geq \lfloor \frac{w-1}{2} \rfloor$. Since $h \leq N-w$, the base $\frac{h}{N-w}$ is at most 1, so lowering the exponent only enlarges the right-hand side. Summing over all $i \in \{0, \dots, w-1\}$ yields

$$\tilde{A}(w, h) \leq w \cdot \frac{w \cdot 6^w}{N} \left(\frac{h}{N-w} \right)^{\lfloor (w-1)/2 \rfloor} = \frac{w^2}{N} \left(\frac{h}{N-w} \right)^{\lfloor (w-1)/2 \rfloor} 6^w.$$

For $D(w)$, write $r = \lfloor (w-1)/2 \rfloor$. Since $\delta \leq 0.30$ and $w \leq 0.01N$, every $h \leq \delta N$ also satisfies $h \leq N-w$, so the previous bound applies throughout the summation range. Using $\sum_{h=1}^{\delta N} h^r \leq \int_1^{\delta N+1} x^r dx = \frac{(\delta N+1)^{r+1}}{r+1}$:

$$\begin{aligned} D(w) &\leq \frac{w^2}{N} \cdot \frac{6^w}{(N-w)^r} \cdot \frac{(\delta N+1)^{r+1}}{r+1} \\ &= \frac{w^2}{r+1} \cdot \frac{\delta N+1}{N} \cdot \left(\frac{\delta N+1}{N-w} \right)^r \cdot 6^w. \end{aligned}$$

D.4 Proof of Lemma 3

Proof (Proof of Lemma 3). Put $\eta_0 = 0.01$. To obtain the explicit choice in Remark 1, set $\delta = \delta_0 = 0.99/(144e^2)$ and assume $N \geq 1/\delta_0$. Then $(\delta_0 N + 1)/N \leq \bar{\delta} := 2\delta_0$. Define

$$a_0 := 6\sqrt{\frac{\bar{\delta}}{0.99}} < e^{-1}.$$

Lemma 2, together with $\lfloor (s-1)/2 \rfloor \geq s/2 - 1$, gives for $s \leq \eta_0 N$

$$D(s) \leq 2s\bar{\delta} \left(\frac{\bar{\delta}}{0.99} \right)^{s/2-1} 6^s = 1.98 s a_0^s.$$

The first accumulator has support only for $s \geq \lceil kw/2 \rceil$. Applying the same lemma to its IOWE gives

$$g(s) := \tilde{A}(kw, s)D(s) \leq \frac{(kw)^2}{N} \left(\frac{s}{N - kw} \right)^{kw/2-1} 6^{kw} \cdot 1.98 s a_0^s.$$

Its logarithmic derivative is at most

$$\frac{kw}{2s} + \log a_0 \leq 1 + \log a_0 < 0.$$

Hence the largest summand is at the left support endpoint, and evaluating the decreasing upper bound at $kw/2$ only enlarges it.

Write $\alpha = kw/N \leq \eta_0$ and

$$b(\alpha) := 6\sqrt{a_0} \sqrt{\frac{\alpha}{2(1-\alpha)}}, \quad b_0 := b(\eta_0) < 1.$$

Combining the preceding bound with $\binom{N/k}{w} \leq (eN/(kw))^w$ and summing at most $\eta_0 N$ intermediate weights yields

$$p(w) \leq N^{O(1)} \left[\frac{e(q-1)}{\alpha} b(\alpha)^k \right]^w.$$

The same second-derivative calculation as in the unparameterized bound shows that its logarithm is convex for $k \geq 9$, so it suffices to check the two endpoints. At $w = 1$, the unsimplified bound is $O_{q,k}(N^{1-\lfloor (k-1)/2 \rfloor}) = o(N^{-2})$ for $k \geq 9$. At $w = \eta_0 N/k$,

$$p(\eta_0 N/k) \leq N^{O(1)} \left[\frac{e(q-1)}{\eta_0} b_0^k \right]^{\eta_0 N/k},$$

which is exponentially small whenever

$$k > \frac{\log(e(q-1)/\eta_0)}{-\log b_0}.$$

These are exactly the two small-weight conditions included in Equation (1).

D.5 Proof of Theorem 4 and Completion of Theorem 2

Proof (Proof of Theorem 4). Put $s_0 = \lceil 0.01N \rceil$ and split the sum at s_0 :

$$\begin{aligned} & \binom{N/k}{w} (q-1)^w \sum_{s=1}^N \tilde{A}(kw, s) D(s) \\ &= \underbrace{\sum_{s=1}^{s_0-1} \binom{N/k}{w} (q-1)^w \tilde{A}(kw, s) D(s)}_{o(1/N^2), \text{ Lemma 3}} + \underbrace{\sum_{s=s_0}^N \binom{N/k}{w} (q-1)^w \tilde{A}(kw, s) D(s)}_{e^{-\Omega(N)}, \text{ Corollary 2}} \end{aligned}$$

The total is $o(1/N^2) + e^{-\Omega(N)} = o(1/N^2)$.

Proof (Proof of Theorem 2 (completion)). Summing (3) over all w :

$$\Pr[\text{dist}(C) < \delta N] \leq \sum_{w=1}^{N/k} \Pr[\dots \mid \text{wt}(x) = w] \leq \sum_{w=1}^{N/k} o(1/N^2) = o(1/N).$$

The large-weight terms ($w \geq 0.01N/k$) are $e^{-\Omega(N)}$ by Corollary 1, and the small-weight terms ($w < 0.01N/k$) are $o(1/N^2)$ by Theorem 4.

E Deferred Proofs for the Concrete Analysis

This appendix gives the counting details behind Theorem 5, a complete proof of Theorem 6, and the parameter-selection procedure.

E.1 Proofs for the Concrete Distance Bound

Proof (Detailed proof of Theorem 5). Fix the h support positions $1 \leq p_1 < \dots < p_h \leq N$ of the accumulator input and define

$$g_0 = p_1 - 1, \quad g_i = p_{i+1} - p_i - 1 \quad (1 \leq i < h), \quad g_h = N - p_h.$$

The support is uniform, so $(g_0, \dots, g_h)$ is a uniform weak composition of $N - h$ into $h+1$ parts. Let $\xi_1, \dots, \xi_h$ be the nonzero input values and $S_r = \sum_{\ell=1}^r \xi_\ell$. For $J \subseteq [h]$, write $j = |J|$ and consider the event $S_r = 0$ for every $r \in J$. Ordering the indices in J partitions the ξ_ℓ into j disjoint nonempty blocks whose sums must vanish. After fixing all but the final value in each block, at most one of the q_0 possible final values satisfies its constraint. Thus

$$\Pr[S_r = 0 \text{ for all } r \in J] \leq q_0^{-j}. \quad (22)$$

The accumulator output is zero on the initial gap of length g_0 , and a zero return at r contributes the support coordinate p_r together with the following gap of length g_r . If $\text{wt}(Ax) \leq d$, then for the actual zero-return set J ,

$$g_0 + \sum_{r \in J} g_r + j \geq N - d.$$

Equivalently, the $h-j$ complementary gaps have total mass at most $d-h+j$. If $d-h+j < 0$, this candidate contributes zero. Otherwise, for a fixed candidate J , condition on their total mass y . The selected $j+1$ gaps have $\binom{N-h-y+j}{j}$ weak compositions, which is at most $\binom{N-h+j}{j}$. Summing the complementary compositions and using the hockey-stick identity gives

$$\sum_{y=0}^{d-h+j} \binom{y+h-j-1}{h-j-1} = \binom{d}{h-j}.$$

The identity also holds at the endpoint $h=j$, under the usual empty-family convention. A union bound over the $\binom{h}{j}$ candidate return sets, followed by (22), yields

$$D(h) \leq \frac{1}{\binom{N}{h}} \sum_{j=0}^h \binom{h}{j} \binom{N-h+j}{j} \binom{d}{h-j} q_0^{-j}. \quad (23)$$

Set $t = h-j$ and use

$$\frac{\binom{N-t}{h-t}}{\binom{N}{h}} = \frac{\binom{h}{t}}{\binom{N}{t}}.$$

Equation (23) becomes

$$D(h) \leq \sum_{t=0}^{\min\{h,d\}} \frac{\binom{h}{t}^2 \binom{d}{t}}{\binom{N}{t}} q_0^{-(h-t)} = U_*(h).$$

It remains to prove the contraction. Denote the t -th summand by

$$u_{h,t} := \frac{\binom{h}{t}^2 \binom{d}{t}}{\binom{N}{t}} q_0^{-(h-t)},$$

and set invalid summands to zero. Direct cancellation gives

$$\frac{u_{h+1,t}}{u_{h,t}} = \frac{(h+1)^2}{q_0(h+1-t)^2}, \quad (24)$$

$$\frac{u_{h+1,t}}{u_{h,t-1}} = \frac{(h+1)^2(d-t+1)}{t^2(N-t+1)} \quad (t \geq 1). \quad (25)$$

Fix $x \in (0,1)$ and put $c = \min\{d, \lfloor x(h+1) \rfloor\}$. For $t \leq c$, Equation (24) is at most $1/[q_0(1-x)^2]$. For $t \geq c+1$, we have $t > x(h+1)$ and

$$\frac{d-t+1}{N-t+1} \leq \frac{d}{N},$$

so Equation (25) is at most $(d/N)/x^2$. Summing the first comparison over $t \leq c$ and the second over $t \geq c+1$ gives

$$U_*(h+1) \leq \left(\frac{1}{q_0(1-x)^2} + \frac{d/N}{x^2} \right) U_*(h).$$

The parent $u_{h,c}$ may appear in both sums, which only loosens the upper bound. Minimization over x gives

$$x_\star = \frac{(d/N)^{1/3}}{q_0^{-1/3} + (d/N)^{1/3}},$$

and substituting $x_\star$ yields $\rho_\star = (q_0^{-1/3} + (d/N)^{1/3})^3$. $\square$

Proof (Detailed proof of Lemma 6). Let $v = (1, r)^T$. From Equation (13),

$$M_a(z)v \leq \lambda(a, z, r)v$$

componentwise. Since $\mathbf{1} \leq C(r)v$, positivity of $M_a(z)$ gives

$$e_0^T M_a(z)^N \mathbf{1} \leq C(r) e_0^T M_a(z)^N v \leq C(r) \lambda(a, z, r)^N e_0^T v = C(r) \lambda(a, z, r)^N.$$

All coefficients of $e_0^T M_a(z)^N \mathbf{1}$ are nonnegative, so

$$\text{coeff}_s(e_0^T M_a(z)^N \mathbf{1}) \leq z^{-s} e_0^T M_a(z)^N \mathbf{1}.$$

Finally, because $0 < a \leq 1$, $\mathbf{1}[H_s \leq R] \leq a^{H_s - R}$. Taking expectations and applying Lemma 5 proves Equation (14). $\square$

Proof (Detailed proof of Theorem 6). For every interval $\mathcal{B}_j = [L_j, R_j]$ and every $h \in \mathcal{B}_j$, Theorem 5 and $\rho_\star < 1$ imply

$$D(h) \leq U_\star(h) \leq U_\star(L_j).$$

Lemma 4 gives $D(h) = 0$ outside $[1, H_\star]$, where $H_\star = \min\{N, 2d\}$. Therefore

$$\begin{aligned} \mathbb{E}[D(H_{kw})] &= \sum_{h=1}^{H_\star} \Pr[H_{kw} = h] D(h) \\ &\leq \sum_{j=1}^J U_\star(L_j) \Pr[H_{kw} \in [L_j, R_j]] \\ &\leq \sum_{j=1}^J U_\star(L_j) \Pr[H_{kw} \leq R_j]. \end{aligned}$$

Use Lemma 6 with $s = kw$ and the parameters $(a_{w,j}, z_{w,j}, r_{w,j})$ in the j -th summand. Multiplication by $\binom{n}{w} q_0^w$ proves Equation (15). Summing over the complete range $1 \leq w \leq n$ and applying Equation (7) proves the failure-probability claim.

For completeness, if one parameter triple (a, z, r) is shared by every $w \in \mathcal{I}$, its factors independent of w can be pulled out of the sum. The remaining finite sum is exactly $\sum_{w \in \mathcal{I}} \Gamma_w(z)$ from Equation (17). Thus the batching step is an exact regrouping of the complete upper bound. $\square$

E.2 Computation and Interval Verification

The computation uses interval partitions of the complete w - and h -ranges. For each distinct h -interval endpoint it evaluates $U_\star(L)$ by a stable summand recurrence. For each pair of intervals it optimizes three positive scalars (a, z, r) and evaluates the corresponding finite T_w sum directly over all integer weights in the interval. A short set of small message weights may instead be evaluated with the exact accumulator IOWE; this improves tightness but is not needed for coverage.

For the screening step, let $S_1 = T(1)$. Summing exact IOWE cells over any selected intermediate-weight subsets gives a lower bound $L_1 \leq S_1$, because every omitted cell is nonnegative. The exact first-accumulator prefix and the $U_\star$ contraction give an upper bound $S_1 \leq U_1$. Directed rounding therefore produces a verified bracket $[L_1, U_1]$. In the unfiltered mode, the lower endpoint can reject a tuple before the full calculation. In the filtered mode, the upper endpoint bounds the rejection probability and expected resampling overhead of Algorithm 2. Full-range certification separately bounds $\sum_{w=2}^n T(w)$ and remains the acceptance test.

The ordinary floating-point pass only chooses the partitions and parameter triples. The verification data consist of these partitions and parameters, together with outward-rounded interval enclosures of every term in Equation (17). The verifier also checks that the exact weights together with the w -intervals partition $[1, n]$, the h -intervals partition $[1, H_\star]$ for $H_\star = \min\{N, 2d\}$, and $\sup \rho_\star < 1$. It returns the sum of the upper interval endpoints. Consequently, numerical overflow, optimizer failure, or an uncovered integer causes refinement or FAIL, never an unsupported numerical answer.

E.3 ParamSelect Algorithm

Algorithm 4 uses RAA-CERTIFY as a feasibility oracle over candidate grids for (k, δ) and returns the accepted tuple minimizing the specified deployment-cost function. For each candidate it separates the certified setup bound into ϵ_1 and $\epsilon_{\geq 2}$ and applies Equation (19) according to the deployment’s filtering policy. Its optimality is therefore relative to the supplied grids and cost function. The proof-error budget is split as $\epsilon_{\text{IOPP}} + \epsilon_{\text{SZ}} \leq \epsilon_{\text{proof}}$, avoiding any dependence between the one-time code-sampling event and per-proof soundness.

Algorithm 4: RAA-PARAMSELECT: setup-time parameter selection.

Input: Codeword length N , field size q , candidate grids for k and δ ;
filtering policy $b_{\text{filter}} \in \{0, 1\}$; budgets ϵ_{setup} and ϵ_{proof}
Output: A cost-minimizing accepted tuple $(k^*, \delta^*, \ell^*, e^*)$, or FAIL

```

1 Choose  $\epsilon_{\text{IOPP}}, \epsilon_{\text{SZ}} > 0$  with  $\epsilon_{\text{IOPP}} + \epsilon_{\text{SZ}} \leq \epsilon_{\text{proof}}$ ;
2  $\text{cost}^* \leftarrow \infty$ ;
3 foreach  $k$  in the candidate repetition grid do
4   foreach  $\delta$  in the candidate distance grid do
5      $(\epsilon_1, \epsilon_{\geq 2}) \leftarrow \text{RAA-CERTIFYBYWEIGHT}(N, q, k, \delta)$ ;
6     if certification fails or  $\epsilon_1 \geq 1$  then
7       continue;
8     if  $b_{\text{filter}} = 1$  then
9        $\epsilon_{\text{mode}} \leftarrow \epsilon_{\geq 2} / (1 - \epsilon_1)$ ;
10    else
11       $\epsilon_{\text{mode}} \leftarrow \epsilon_1 + \epsilon_{\geq 2}$ ;
12    if  $\epsilon_{\text{mode}} > \epsilon_{\text{setup}}$  then
13      continue;
14     $\ell \leftarrow \lceil \ln(1/\epsilon_{\text{IOPP}}) / [-\ln(1 - \delta/3)] \rceil$ ;
15     $e \leftarrow \lceil \log_q(N/\epsilon_{\text{SZ}}) \rceil$ ;
16     $\text{cost} \leftarrow \text{DEPLOYMENTCOST}(N, k, \ell, e, b_{\text{filter}})$ ;
17    if  $\text{cost} < \text{cost}^*$  then
18      Update  $(k^*, \delta^*, \ell^*, e^*, \text{cost}^*)$ ;
19 if  $\text{cost}^* = \infty$  then
20   return FAIL
21 return  $(k^*, \delta^*, \ell^*, e^*)$ 

```

After parameter selection, the unfiltered mode samples one concrete code description directly. The filtered mode instead invokes Algorithm 2 and checks every weight-one message. All certified quantities in the selector, including ϵ_1 , $\epsilon_{\geq 2}$, and ϵ_{mode} , are evaluated with outward rounding.

F Deferred Details: IOPP-Level Optimizations and FHE Costs

This appendix contains the deferred material from Sections 6.1–6.4: detailed per-step FHE cost analyses (Sections 6.2 and 6.3) and the full step-1/step-2 description of ciphertext repacking (Section 6.4).

F.1 Per-Step FHE Cost Tables for the IOPP, PIOP, and PCS

We tabulate the per-step FHE costs that underlie the asymptotic claims of Sections 6.1–6.5. Table 4 reports the per-row encoding cost of our RAA code against an RS code under the low-depth 2D-NTT trick introduced by HELIOPOLIS [4] and used by PHALANX.

Table 4. Per-row FHE encoding cost comparison (m = row length, rate ρ). This is the cost that the IOPP commit phase incurs.

Code	Pt-Ct Mults	Depth	Scaling
RS (2D-NTT)	$k(m/\rho)^{1+1/k}$	k	$O(m^{1+1/k})$
RAA (ours)	$2m/\rho$	1	$O(m)$

PIOP per-step costs (expanded from Section 6.2). The Spartan PIOP for R1CS has three phases. (i) The encoded R1CS matrices A, B, C are public; computing $M_a = Az$, $M_b = Bz$, $M_c = Cz$ in the encrypted setting costs $O(n)$ pt-ct multiplications at depth 0. (ii) The first sumcheck (degree 3, $\log n$ rounds) over $\tilde{eq}(r_x, x)(\widetilde{M_a(x)M_b(x)} - \widetilde{M_c(x)})$: per round, the sum-computation contributes ct-ct multiplications geometrically (round i does $n/2^i$ ct-ct mults), summing to $O(n)$ ct-ct mults total at depth ≤ 2 ; the depth-controlled folding contributes $O(n)$ mult_plain per round at depth 1, summing to $O(n \log n)$ mult_plain total. (iii) The second sumcheck (degree 2, $\log n$ rounds) over $\tilde{z}(y)\widetilde{M_{r'_x}}(y)$: the integrand has $\widetilde{M_{r'_x}}$ as a plaintext multilinear polynomial (the verifier evaluates it on its own), so the per-round work is $O(n/2^i)$ mult_plain for the sum-computation plus $O(n)$ mult_plain for the depth-controlled folding, totalling $O(n \log n)$ mult_plain at depth 1 with no ct-ct multiplications. Total PIOP cost: $O(n)$ ct-ct multiplications, $O(n \log n)$ mult_plain, FHE depth $O(1)$.

PCS per-step costs (expanded from Section 6.3). The commit phase invokes IOPP.Commit at cost $O(N)$ pt-ct ops, depth 1. The two LinComb queries (one for proximity, one for evaluation) are $O(N)$ pt-ct ops each at depth 0. Total: $O(N)$ FHE operations at depth 1, dominated by the IOPP’s commit phase, strictly linear with no $\log N$ factor.

F.2 Ciphertext Repacking (Step 1 and Step 2)

The problem repacking solves. The IOPP LinComb response is the length- $\hat{n}$ vector $u^* = (u_1^*, \dots, u_{\hat{n}}^*)$ with $u_j^* = \sum_{i=1}^B \gamma_i f_{i,j}$, where $f_{i,j}$ is the encoded value of row i at column j . Because the data is SIMD-packed *by column* (column j occupies one ciphertext, with the B rows in its B slots), each scalar u_j^* is initially produced inside its own column ciphertext as a cross-slot sum, so u^* is spread across M ciphertexts, each using 1 of its B slots and wasting the other $B - 1$. Transmitting all M ciphertexts is hugely redundant (~ 196 MB at $N = 2^{21}$). *Repacking* (Algorithm 5) gathers the M scalars into a single packed ciphertext (when $M \leq B$, true for all $N \leq 2^{25}$), an M -fold reduction in transmitted ciphertexts.

Cost: $2\hat{n}$ pt-ct multiplications (the γ scaling and the $\hat{n}$ masks) $+ \hat{n} \log_2 B$ rotations $+ \hat{n} - 1$ ct-ct additions, all at *zero additional multiplicative depth* (rotations and pt-ct multiplications consume no ct-ct level). Repacking consolidates the u^* response from $\hat{n}$ column ciphertexts down to 1, so the proof is

Algorithm 5: REPACK: consolidate the LinComb response u^* into one ciphertext

Input: Column ciphertexts $\text{ct}_{j \in [\hat{n}]}$, where ct_j packs the B per-row values $(f_{1,j}, \dots, f_{B,j})$ of encoded column j ; row-combination coefficients $\gamma = (\gamma_1, \dots, \gamma_B)$ as one plaintext; SIMD batch B (assume $M \leq B$).

Output: One packed ciphertext ct^* holding $(u_1^*, \dots, u_{\hat{n}}^*)$,

$$u_j^* = \sum_{i=1}^B \gamma_i f_{i,j}.$$

// Step 1 (rotate-and-sum): collapse each column to its scalar u_j^* , replicated across all B slots

```

1 for  $j = 1, \dots, \hat{n}$  do
2    $\text{ct}_j \leftarrow \gamma \cdot \text{ct}_j$ ; /* slot-wise pt-ct mult; slot  $i$  now holds  $\gamma_i f_{i,j}$  */
3   for  $t = 0, \dots, \log_2 B - 1$  do
4      $\text{ct}_j \leftarrow \text{ct}_j + \text{Rot}(\text{ct}_j, 2^t)$ ; /* rotate-and-add; after the loop every slot =  $u_j^*$  */
// Step 2 (mask-and-add): copy each  $u_j^*$  into distinct slot  $j$  of one output ciphertext
5  $\text{ct}^* \leftarrow 0$ ;
6 for  $j = 1, \dots, \hat{n}$  do
7    $\text{ct}^* \leftarrow \text{ct}^* + \mathbf{e}_j \cdot \text{ct}_j$ ; /*  $\mathbf{e}_j$  = one-hot plaintext mask on slot  $j$ ; isolates  $u_j^*$  into slot  $j$  */
8 return  $\text{ct}^*$ ;
```

$(\ell + 1) \times 128 \text{ KB} \approx 105.5 \text{ MB}$ at $N = 2^{22}$ ($\ell^* = 843$), versus $(\ell + \hat{n}) \times 128 \text{ KB}$ without repacking, a $\sim 2.2\times$ reduction; the repack step itself adds only $\sim 33 \text{ ms}$ of prover time (measured, Section 7).

F.3 Complete Blind PCS Protocol

For completeness we spell out the full blind PCS for μ -variate multilinear evaluation, assembling the components of Section 6.3: the FHE-friendly IOPP construction of Algorithm 3, the tensor decomposition of the evaluation, and the ciphertext repacking of Algorithm 5. The scheme is designated-verifier: the verifier $\mathcal{V}$ holds the FHE secret key sk , the prover $\mathcal{P}$ holds only the evaluation key. Completeness, binding, and knowledge soundness are proved in Appendix B.

Commit costs $O(N)$ FHE operations at depth 1 and dominates; Eval adds two depth-0 linear combinations, ℓ Merkle openings, and one repacking at no extra depth. The verifier performs no homomorphic arithmetic: it decrypts the $O(\ell)$ opened ciphertexts and the two responses, then encodes and checks entirely in plaintext. A single Eval has knowledge-soundness error $\epsilon_{\text{PCS}} \leq (1 - \delta/3)^\ell + N/q^e$.

Algorithm 6: Blind PCS for μ -variate multilinear evaluation, designated-verifier

Input: encrypted coefficients $\{\llbracket \tilde{f}(x) \rrbracket\}_{x \in \{0,1\}^\mu}$, $N = 2^\mu$, viewed as an $r \times m$ matrix with rows $\llbracket f_i \rrbracket$ and $r = 2^{\lceil \mu/2 \rceil}$; rate- ρ Generalized RAA code C of relative distance $\geq \delta$, encoder E_C ; evaluation point $z \in \mathbb{F}_q^\mu$. With $s = \lceil \mu/2 \rceil$, the tensors $q_1 = \bigotimes_{i < s} (1 - z_i, z_i) \in \mathbb{F}_q^r$, $q_2 = \bigotimes_{i \geq s} (1 - z_i, z_i) \in \mathbb{F}_q^m$ give $\tilde{f}(z) = \langle q_2, \sum_i q_{1,i} \tilde{f}_i \rangle$.

// Setup

- 1 Audit the chosen code parameters by Algorithm 1; fix random oracle H , query count ℓ , extension degree e ;

// Commit

- 2 $\text{com} \leftarrow \text{IOPP.COMMIT}(\{\llbracket f_i \rrbracket\}_{i \in [r]})$; /* Algorithm 3: RAA-encode each row, Merkle by column */
- 3 ;

// Eval at z

- 4 $\mathcal{V} \rightarrow \mathcal{P}$: $\gamma \in \mathbb{F}_{q^e}^r$ and columns $I \subset [\hat{n}]$, $|I| = \ell$; /* $(\gamma, I) = H(\text{com}, z)$ under Fiat-Shamir */
- 5 ;
- 6 $\mathcal{P} \rightarrow \mathcal{V}$: $\llbracket u \rrbracket \leftarrow \text{IOPP.LINCOMB}(\gamma)$; $\llbracket f'' \rrbracket \leftarrow \text{IOPP.LINCOMB}(q_1)$, repacked by Algorithm 5; column openings $\{\llbracket c_{\cdot,j} \rrbracket\}_{j \in I}$ with Merkle paths;
- 7 $\mathcal{V}$ verifies the Merkle paths, then decrypts $u \leftarrow \text{Dec}_{\text{sk}}(\llbracket u \rrbracket)$, $f'' \leftarrow \text{Dec}_{\text{sk}}(\llbracket f'' \rrbracket)$, and $c_{i,j} \leftarrow \text{Dec}_{\text{sk}}(\llbracket c_{i,j} \rrbracket)$, $j \in I$; /* all checks below are in plaintext */
- 8 ;
- 9 $\mathcal{V}$ checks $E_C(u)_j = \sum_i \gamma_i c_{i,j}$ and $E_C(f'')_j = \sum_i q_{1,i} c_{i,j}$ for all $j \in I$, then outputs $y \leftarrow \langle q_2, f'' \rangle$ as $\tilde{f}(z)$;

G Detailed Related Work Survey

This appendix expands the brief overview in Section 2 into a detailed technical comparison. We organize the literature into three threads: (i) distance analysis of accumulator-based codes (the analytical lineage our work belongs to), (ii) blind PCS and verifiable FHE (the application setting), and (iii) coding-based PCS more generally (the broader design space).

G.1 Distance Analysis of Accumulator-Based Codes

The distance of accumulator-based codes over the binary field has been studied for over two decades [13, 21, 12]. Pfister et al. [21] established that L -fold accumulation with $L \geq 2$ suffices for linear minimum distance over $\mathbb{F}_2$; Blaze [12]

leveraged this to construct efficient interleaved-code-based SNARKs with linear-time provers. These analyses are tight for $\mathbb{F}_2$ but do not transfer to non-binary fields, because their core combinatorial identities exploit binary-specific cancellations (e.g. XOR of two ones equals zero).

Non-binary in the $n \gg q$ regime. Three threads have studied non-binary accumulator codes in the regime where the block length n greatly exceeds the field size q , but each falls short of a rigorous closed-form distance bound:

- *Code family introduction.* Yang [28] introduced weighted nonbinary repeat-accumulate codes and gave simulation-based evidence on the AWGN channel; no minimum-distance theorem is stated.
- *Approximate IOWE for performance analysis.* Kim, Cheun, and Lim [20] derived an approximate input-output weight enumerator (IOWE) for the nonbinary accumulator and used it to compute approximate ML decoding thresholds; the bound is asymptotic and does not yield a concrete failure probability for finite N .
- *Asymptotic distance with numerical verification.* Rosnes and Graell i Amat [25,17] give the first formal asymptotic distance theorem, proving ([25, Thm. 4]) that the symbol-wise minimum distance grows linearly with the block length *conditional on the strict positivity of a spectral-shape-function threshold* ρ_0 . The verification of $\rho_0 > 0$ ([25, Thm. 5]) is performed by *numerically solving a non-convex optimization over a high-dimensional simplex*, with the result reported only at the discrete set of field sizes $q \in \{3\} \cup \{2^l : 2 \leq l \leq 25\}$. For $q > 2^{25}$ positivity is asserted by graphical extrapolation (“from the figures we observe that the result will also hold for larger values of q ” [25, §IV]). The framework provides neither closed-form q -dependence of ρ_0 , nor a concrete failure probability for finite N , nor explicit constants applicable to a target (N, q, k, δ) .

Asymptotically large-field regime. A separate line of work targets the regime $q = \omega(\text{poly}(n))$, where the field size grows super-polynomially. Block et al. [8] gave the first rigorous large-field analysis for Expander-Accumulate codes. Akhiani and Zhang [2] later obtained failure probability $\tilde{O}(N^{-\eta}) + O(1/(q-1))$ for RAA codes using generating functions. Their bound is tighter than ours when q grows super-polynomially. For fixed q , however, the $O(1/(q-1))$ term remains. It arises from partitioning the IOWE sum and retaining the nonzero-symbol factor on only one part. This term prevents the bound from certifying the fixed plaintext modulus used by FHE.

Quasi-Abelian codes (concurrent). A very recent ePrint [22] introduces Quasi-Abelian (QA) codes over group rings $\mathbb{F}_p[\mathbb{Z}_2^n]$ and obtains a concrete-regime distance bound that approaches the Gilbert-Varshamov line with gap $(1 + \log_p N)/c$, where c is the inverse code rate. For p much larger than N (e.g. a 128-bit prime with $N \leq 2^{40}$), the gap is small and they obtain very strong concrete distances (e.g. $\delta \geq 0.41$ at rate $1/2$). As the authors explicitly note, however, “it

remains open whether QA code is asymptotically good for fixed index c and field size p , as our lower bound is meaningless when $|G| = N \rightarrow \infty$." Concretely, when p is the small FHE plaintext prime ($q = 2^{16} + 1$ in our setting), the $\log_p N/c$ gap is already $1.25/c$ at $N = 2^{20}$ and grows without bound, so QA codes provide no usable distance guarantee in the FHE-plaintext regime that motivates our work. In addition, QA encoding uses an $O(\log N)$ -stage Walsh–Hadamard Transform, inheriting the same multiplicative-depth limitation as NTT-style transforms, so it does not resolve the complexity–depth dilemma.

Our position. The technical core of our proof, Lemma 2, gives an elementary combinatorial upper bound on $\tilde{A}(w, h)$. It avoids both the spectral-shape analysis of [25, 17] and the partitioned generating-function analysis of [2]. The proof uses $(q - 1)^a (q - 2)^i / (q - 1)^w \leq 1$ uniformly over the complete IOWE sum and then bounds the remaining combinatorial terms directly. Consequently, it has no irreducible $O(1/(q - 1))$ term. The prior decomposition is tighter when q grows, while ours is tighter for fixed q . The latter is the deployment regime for blind PCS over BGV/BFV ciphertexts.

G.2 Blind PCS and Verifiable FHE

The development of blind PCS is closely tied to the adaptation of interactive proofs to the FHE domain. Fiore et al. [14] and Bois et al. [9] studied verifiable computation over encrypted data, establishing the theoretical foundations. The concept of performing proximity testing over encrypted values can be traced to HELIOPOLIS [4], which provided the first study on applying the FRI (Reed–Solomon IOPP) protocol to FHE ciphertexts, though their security model is limited to freshly encrypted values.

Constant-depth full SNARKs. Building on the low-depth 2D-NTT evaluation of Reed–Solomon encoding introduced by HELIOPOLIS [4], PHALANX [30] proposed the first constant-depth full SNARK for the VCoED setting. However, both its prover time and proof size scale super-linearly as the circuit grows: the prover incurs $O(kN^{1+1/k})$ FHE operations.

Linear-prover blind PCS with $O(\log n)$ depth. LAMINATE [23] addresses the proof-size explosion by adapting the GKR protocol and incorporating folding-based techniques like blind BaseFold, achieving $O(N \log N)$ prover, but its outer PCS still incurs $O(N \log N)$ FHE operations and inherits $O(\log N)$ multiplicative depth from the underlying FFT.

Concurrent application of RAA codes in the plaintext domain. A parallel line of work by Akhiani et al. [1] applies non-binary RAA codes in the *plaintext* domain for outsourced SNARKs. Our work targets the *encrypted* (blind) setting, which additionally requires constant multiplicative depth and SIMD-compatible operations, and integrates the code into a full PCS with constant-depth ciphertext repacking.

Our position. No prior blind PCS simultaneously achieves $O(1)$ FHE multiplicative depth and $O(N)$ prover work. Our PCS is the first to resolve the “complexity-depth dilemma” at the PCS layer (identified explicitly by [23]), by replacing the underlying code with the Generalized RAA structure analyzed in this paper. At the SNARK level, our composition with a depth-controlled Spartan PIOP retains $O(N)$ ct-ct multiplications and $O(1)$ FHE depth, with an additional $O(\log N)$ factor in total work arising from the depth-controlled sumcheck folding (Section 6.2), asymptotically matching LAMINATE’s $O(N \log N)$ total work at constant depth, while supporting arbitrary R1CS rather than layered uniform circuits.

G.3 Coding-Based PCS

We categorize coding-based PCS into three families based on the underlying code.

Reed–Solomon (RS) based constructions (FRI [7], BaseFold [29], Ligero [3], DeepFold [18], WHIR [5]) exploit the optimal distance of RS codes but inherently require $O(n \log n)$ encoding via NTT. In the blind setting, the logarithmic multiplicative depth of NTTs (or the complexity explosion of constant-depth grouped NTTs) poses a significant bottleneck.

Linear-time encodable constructions such as Brakedown [16] and Orion [27] replace RS codes with expander-based codes, achieving $O(n)$ prover time and $O(\log^2 n)$ proof size. While asymptotically optimal in the plaintext domain, their internal algebraic structures involve complex bit-level operations or high-depth circuits, making them impractical for FHE.

Accumulator-based constructions (Blaze [12], EA codes [8]) leverage prefix-sum operations for $O(1)$ depth encoding, but existing work is limited to binary fields (Blaze) or requires super-linear encoding (EA, $O(n \log n)$). Our Generalized RAA code is the first to simultaneously achieve $O(n)$ encoding, $O(1)$ depth, and non-binary field support. Table 1 (in Section 2) provides a comparative summary.